

McKnight Road Retail Development



Traffic Impact Study

Prepared September 16, 2026 · 40.5250, -80.0130

Simple Impact Studies — sample draft

Traffic Impact Study

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1.0 EXECUTIVE SUMMARY

This Traffic Impact Study (TIS) evaluates the transportation impacts of the proposed McKnight Road Retail Development located within Pittsburgh MSA, Pennsylvania. Analysis follows PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual (2019) and applicable PennDOT standards. The study area encompasses 36 intersections within a 2.00-mile radius from the site, analyzed for Land Use 820 (Shopping Center / Retail) with a development size of 200 1,000 sqft GLA.

Key Findings:

- 10 intersections are projected to drop one or more LOS grades under Build conditions.
- 4 intersections operate at LOS E or F under Build conditions — mitigation analysis is required per PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual.

36 INTERSECTIONS	10 LOS DROPS	4 AT LOS E/F	17.3s WORST Δ DELAY
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2.0 PROJECT DESCRIPTION

Project name	McKnight Road Retail Development
Land use	820 — Shopping Center / Retail
Development size	200 1,000 sqft GLA
Site coordinates	40.5250°N, 80.0130°W
Jurisdiction	Pittsburgh MSA
State	Pennsylvania
State DOT	Pennsylvania Department of Transportation (PennDOT)
Opening year	2028
Study type	Traffic Impact Study (TIS)
Controlling document	PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual

Site plan, surrounding land use, and access configuration are dependent on the final site plan and are not auto-generated by this tool. Final submittal should incorporate a site plan figure per controlling guidelines.

3.0 METHODOLOGY

3.1 Governing Documents and Controlling Agency

This study is prepared in conformance with PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual (2019). The reviewing agency is the Pennsylvania Department of Transportation (PennDOT).

3.2 Pre-Application Methodology Meeting

A pre-application methodology meeting with PennDOT is required before data collection. PennDOT Pub 282 requires a scoping meeting with the PennDOT District Traffic Unit before data collection. A written scope confirmation must be received before commencing counts.

3.3 Study Area and Analysis Periods

Study area: all signalized and unsignalized intersections and roadway segments within 2.00 miles of the site that receive $\geq 10\%$ of project traffic. Primary analysis periods: weekday AM peak hour and weekday PM peak hour. Additional periods (Saturday peak, midday) are required only where special characteristics warrant, consistent with PennDOT Publication 282 — Guidelines for the Development

and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual.

3.4 Analysis Scenarios

Three scenarios evaluated at each study location:

- Scenario 1 — Existing Conditions: field-collected counts for current peak-hour volumes and geometric conditions.
- Scenario 2 — Future No-Build (2028): background growth applied at 1.50%/yr compound over 2 years, without project traffic.
- Scenario 3 — Future Build (2028): No-Build volumes plus distributed project-generated trips.

3.5 Trip Generation

Trip generation is calculated using Public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), as used in Pennsylvania screening practice. PennDOT Pub 282 explicitly references ITE. The published rate or equation for Land Use Code 820 (Shopping Center / Retail) is applied to a development size of 200 1,000 sqft GLA. Pass-by credit of 30% and internal capture of 0% are applied per standard screening procedures and agreed in the methodology meeting.

3.6 Background Growth Rate

Background traffic growth: 1.50% per year (PennDOT traffic count data (PENNDOT Count Data — online count station database); DVRPC (Philadelphia metro), SPC (Pittsburgh metro), LVPC (Lehigh Valley), or applicable MPO travel demand model.). Source: Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. No measured growth rate is wired for this region, so there is no measured value to compare the override against. The applied rate is not derived from measured data; its basis must be stated by the preparer..

3.7 Level of Service Standards

Acceptable LOS per PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual: LOS D in urbanized areas; LOS C in rural areas. LOS D in urbanized areas; LOS C in rural areas per PennDOT Pub 282. Pennsylvania has one of the most detailed published TIS guidance documents in the US (Pub 282) — follow it closely. The controlling standard evaluates LOS per the Highway Capacity Manual (current edition) via HCS and/or Synchro; this screening applies the openly-published Webster/Akçelik signalized model, superseded by those tools at submittal. Synchro/SimTraffic and HCS standard per PennDOT Pub 282. PennDOT District 6 (Philadelphia) may require VISSIM for signalized corridor studies.

4.0 EXISTING CONDITIONS

4.1 Roadway Network

The existing roadway network within the study area was inventoried to document functional classification, number of travel lanes, posted speed limits, traffic control, and access management characteristics. Data sources: PennDOT functional classification maps, field observations, and Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. No measured growth rate is wired for this region, so there is no measured value to compare the override against. The applied rate is not derived from measured data; its basis must be stated by the preparer..

4.2 Traffic Volumes

Traffic counts were collected during weekday AM (7–9 AM) and PM (4–6 PM) peak periods. Count data represents typical weekday conditions, adjusted to peak-season or peak-annual-day volumes per PennDOT count methodology. AADT base: — vehicles per day (2028).

Base AADT	— vpd
Growth rate applied	1.50%/yr

Growth source

Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. No measured growth rate is wired for this region, so there is no measured value to compare the override against. The applied rate is not derived from measured data; its basis must be stated by the preparer.

Count collection period

Weekday AM + PM peak (field-collected)

4.3 Existing Intersection Operations

Intersection	Control	Dist (mi)	Exist LOS	Delay (s/veh)
Near McKnight Road	Signal	0.38	C	30.3
Babcock Boulevard & Rochester Road	Signal	0.40	B	14.9
Siebert Road & McKnight Road	Signal	0.41	B	12.4
McKnight Road & Braunlich Drive	Signal	0.44	C	26.1
Babcock Boulevard & Siebert Road	Signal	0.44	B	15.2
McKnight Road & Hilands Place	Signal	0.58	C	21.1
Cemetery Lane & Babcock Boulevard	Signal	0.66	D	39.3
Glenmore Avenue & Rochester Road	Signal	0.71	B	11.8
Babcock Boulevard & Evergreen Road	Signal	0.85	B	11.0
McKnight Road & Patrick Place	Signal	0.87	C	34.3
Perry Highway & West View Park Drive	Signal	1.07	B	14.0
Center Avenue & Perry Highway	Signal	1.08	B	14.4
Perry Highway & Chalfonte Avenue	Signal	1.12	B	13.6
Perry Highway & Highland Avenue	Signal	1.14	B	13.5
McKnight Road & Ross Park Mall Drive	Signal	1.17	C	28.3
Bellevue Road & Perry Highway	Signal	1.18	B	15.7
Perry Highway & Rochester Road	Signal	1.19	B	10.3
McKnight Road & McKnight Circle	Signal	1.19	C	28.2
Perry Highway & Perry Lane	Signal	1.21	B	15.6
Center Avenue & West View Park Drive	Signal	1.26	B	10.7
McKnight Road & Nelson Run Road	Signal	1.29	A	8.0
Babcock Boulevard & Thompson Run Road	Signal	1.39	B	19.0
McKnight Road & Johnanna Drive	Signal	1.45	D	43.0
Perrysville Avenue & Perrysville Road	Signal	1.46	B	15.9
Perry Highway & Good Lane	Signal	1.56	B	13.8
Perrysville Avenue & Cemetery Lane	Signal	1.62	D	36.3
Perry Highway & Lindley Lane	Signal	1.63	B	14.8
Three Degree Road & Babcock Boulevard	Signal	1.68	C	22.1
Peoples Road & Babcock Boulevard	Signal	1.76	B	17.2
Perrysville Avenue & West View Avenue	Signal	1.78	F	137.8
McKnight Road & McIntyre Road	Signal	1.78	D	37.4
Perry Highway & Graham Road	Signal	1.81	B	11.7
Sewickley Oakmont Road & Lombardy Drive	Signal	1.85	A	9.6

Intersection	Control	Dist (mi)	Exist LOS	Delay (s/veh)
Perrysville Avenue & Nelson Run Road	Signal	1.89	A	9.5
Babcock Boulevard & Browns Lane	Signal	1.90	C	32.7
Babcock Boulevard & Roderick Drive	Signal	1.95	D	37.9

5.0 TRIP GENERATION ANALYSIS

Trip generation for the proposed McKnight Road Retail Development was estimated using public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for Land Use Code 820 (Shopping Center / Retail). A 30% pass-by credit and 0% internal capture credit are applied per methodology meeting agreement, yielding the following net new external trips:

Land Use Code	820 — Shopping Center / Retail
Development size	200 1,000 sqft GLA
Weekday daily (gross)	16,000
AM peak hour (net new)	640 trips (390 in / 250 out)
PM peak hour (net new)	1,600 trips (768 in / 832 out)
Pass-by credit applied	30%
Internal capture applied	0%
Transit/mode reduction	Not applied

5.1 Trip Generation by Period

Period	Gross Trips	Pass-by	Int. Capture	Net External	In	Out
AM Peak Hour	640	48	0	455	277	178
PM Peak Hour	1,600	480	0	860	413	447
Saturday Midday	1,760	132	0	1,250	625	625
Daily Total	16,000	1,200	0	11,364	5,682	5,682

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross – pass-by – internal-capture) × auto-mode share (76.9% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

Trip Distribution by Time of Day

Estimated within-day distribution of the 16,000 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

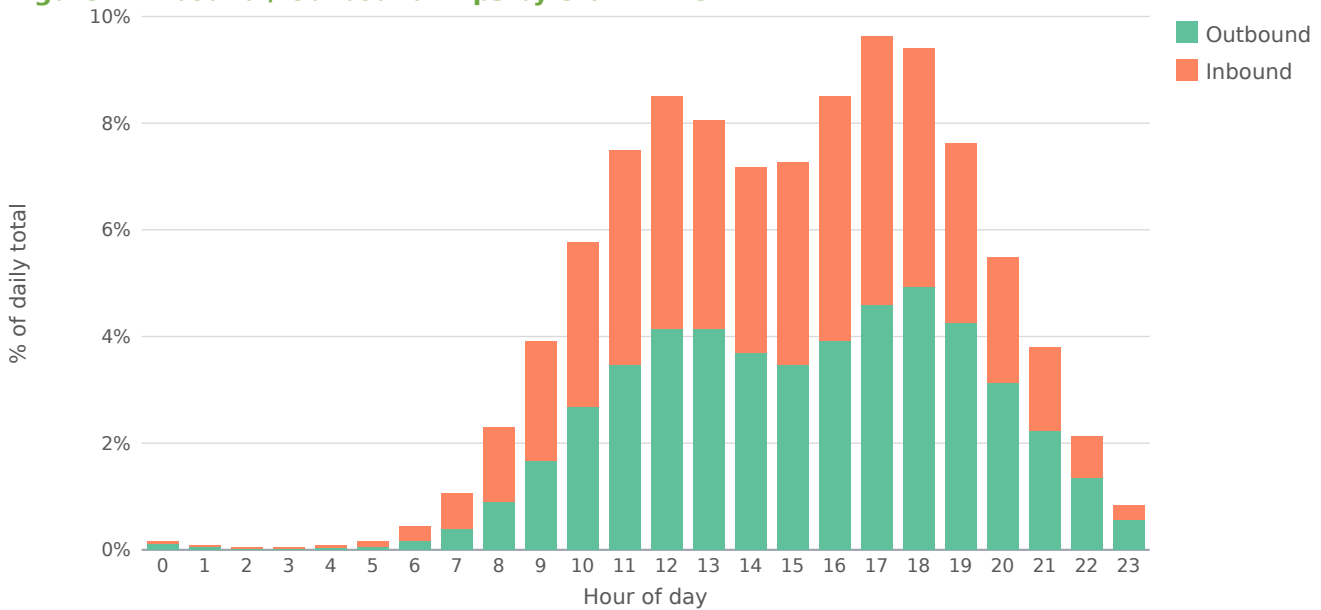
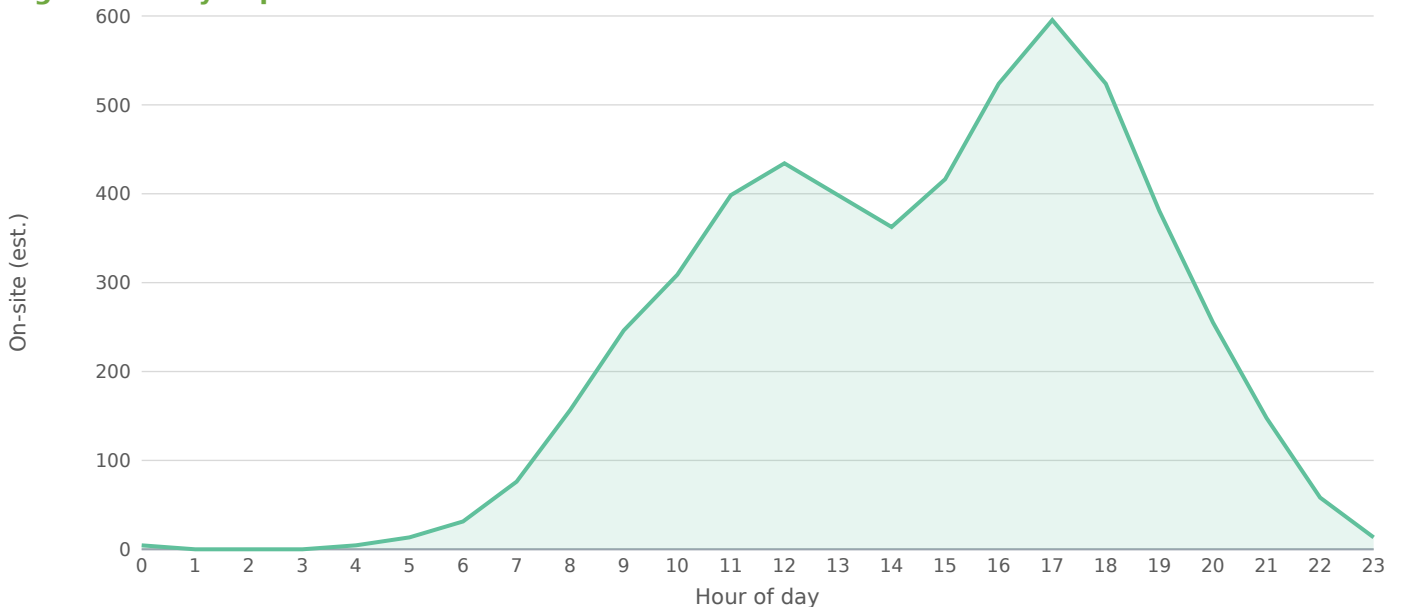


Figure — Daily Trip Accumulation



Peak on-site accumulation ~595 at 17:00, from 16,000 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

6.0 TRIP DISTRIBUTION AND ASSIGNMENT

Project-generated trips are distributed to the roadway network based on the existing directional distribution observed at the study-area count locations, proximity to the site access points, and the regional travel-demand model (PennDOT traffic count data).

Distribution method

Existing traffic directional distribution + engineering judgment

Assignment method

Site-access-proportional; confirmed at methodology meeting

Directional distribution percentages and trip assignment maps should be verified at the methodology meeting with the reviewing agency and attached as Appendix B.

6.1 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = $M / (d^1 \cdot d_{\text{site}})$); mass basis: intersection

through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	16%
Southeast (ESE + SSE)	21%
Southwest (SSW + WSW)	29%
Northwest (WNW + NNW)	33%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Near McKnight Road	ENE	0.38	1,657	0.1	5.19%
McKnight Road & Braunlich Drive	ESE	0.44	1,657	0.1	5.08%
McKnight Road & Hilands Place	ESE	0.58	1,657	0.0	4.85%
Cemetery Lane & Babcock Boulevard	SSE	0.66	1,607	0.0	4.58%
Perrysville Avenue & West View Avenue	SSW	1.78	2,202	0.0	4.41%
McKnight Road & Ross Park Mall Drive	NNE	1.17	1,791	0.0	4.34%
McKnight Road & Patrick Place	NNE	0.87	1,624	0.0	4.33%
McKnight Road & Johnanna Drive	NNW	1.45	1,791	0.0	3.97%
McKnight Road & McKnight Circle	NNW	1.19	1,618	0.0	3.90%
McKnight Road & McIntyre Road	NNW	1.78	1,791	0.0	3.59%
Babcock Boulevard & Rochester Road	SSW	0.40	1,111	0.0	3.45%
Perrysville Avenue & Cemetery Lane	SSW	1.62	1,635	0.0	3.44%
Babcock Boulevard & Siebert Road	NNW	0.44	1,111	0.0	3.40%
Three Degree Road & Babcock Boulevard	NNW	1.68	1,618	0.0	3.35%
Babcock Boulevard & Thompson Run Road	ESE	1.39	1,397	0.0	3.16%
Babcock Boulevard & Browns Lane	NNW	1.90	1,618	0.0	3.13%
Babcock Boulevard & Roderick Drive	NNW	1.95	1,618	0.0	3.08%
Perry Highway & West View Park Drive	WSW	1.07	1,007	0.0	2.52%
Center Avenue & Perry Highway	WSW	1.08	1,007	0.0	2.51%
Perry Highway & Chalfonte Avenue	WSW	1.12	1,007	0.0	2.48%

36 study-area zone(s) (top 20 by trip share shown). Screening-grade distribution; not a calibrated regional model.

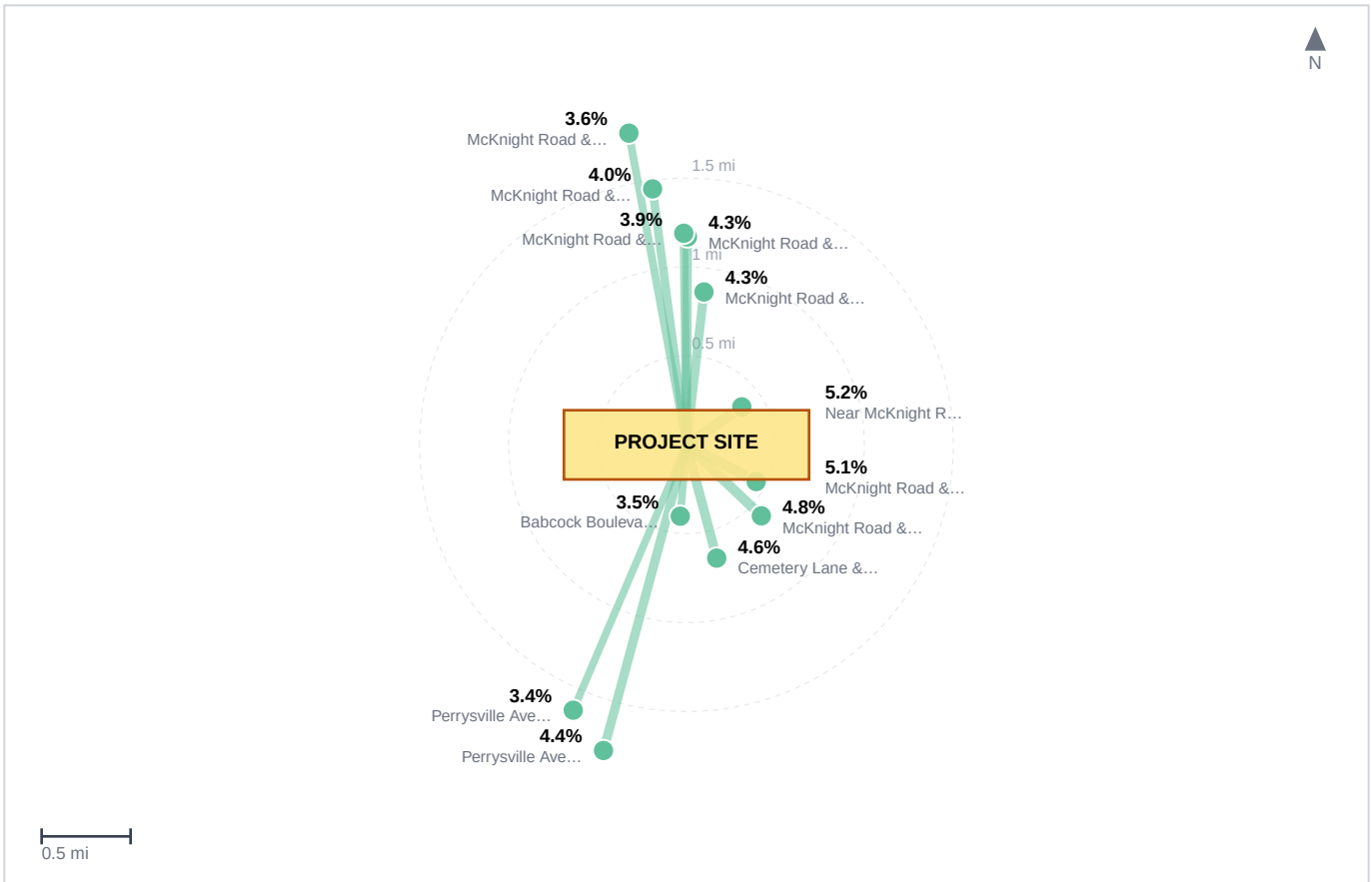
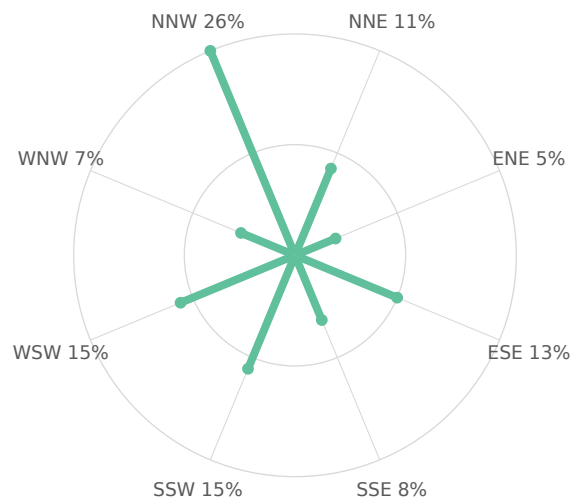


Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

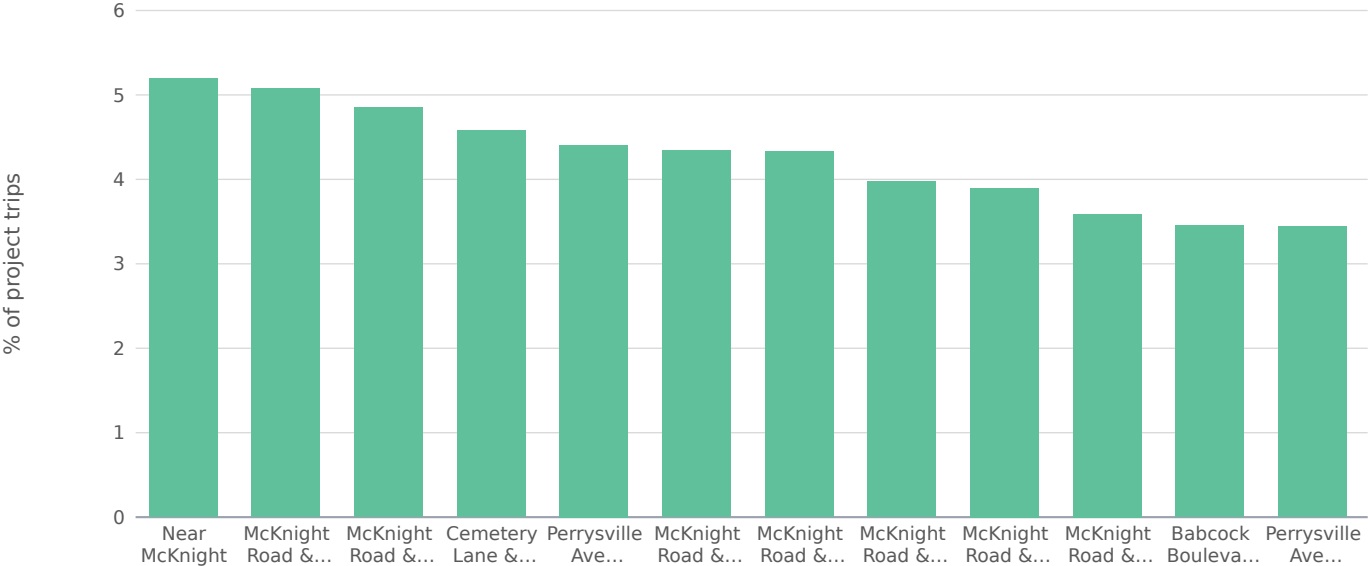
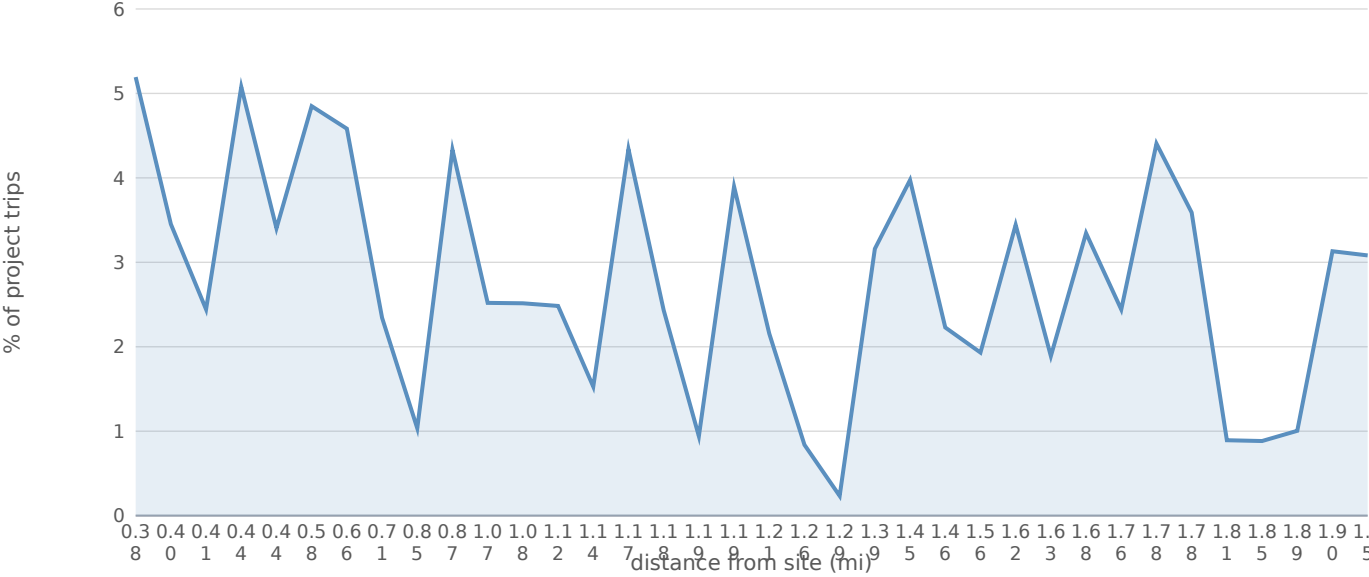


Figure — Trip Share vs. Distance from Site (Gravity Decay)



6.2 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Near McKnight Road	ENE	0.38	69	130	4.8%
Babcock Boulevard & Rochester Road	SSW	0.40	66	112	4.2%
Siebert Road & McKnight Road	NNE	0.41	156	232	8.6%
McKnight Road & Braunlich Drive	ESE	0.44	50	94	3.5%
Babcock Boulevard & Siebert Road	NNW	0.44	99	205	7.6%
McKnight Road & Hilands Place	ESE	0.58	25	47	1.7%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Cemetery Lane & Babcock Boulevard	SSE	0.66	66	112	4.2%
Glenmore Avenue & Rochester Road	WSW	0.71	41	78	2.9%
Babcock Boulevard & Evergreen Road	SSE	0.85	6	12	0.4%
McKnight Road & Patrick Place	NNE	0.87	37	93	3.5%
Perry Highway & West View Park Drive	WSW	1.07	2	4	0.1%
Center Avenue & Perry Highway	WSW	1.08	2	4	0.1%
Perry Highway & Chalfonte Avenue	WSW	1.12	2	3	0.1%
Perry Highway & Highland Avenue	WSW	1.14	2	3	0.1%
McKnight Road & Ross Park Mall Drive	NNE	1.17	37	93	3.5%
Bellevue Road & Perry Highway	WSW	1.18	1	2	0.1%
Perry Highway & Rochester Road	WNW	1.19	41	78	2.9%
McKnight Road & McKnight Circle	NNW	1.19	101	151	5.6%
Perry Highway & Perry Lane	WNW	1.21	1	2	0.1%
Center Avenue & West View Park Drive	WSW	1.26	1	2	0.1%
McKnight Road & Nelson Run Road	SSE	1.29	56	140	5.2%
Babcock Boulevard & Thompson Run Road	ESE	1.39	33	49	1.8%
McKnight Road & Johnanna Drive	NNW	1.45	37	93	3.5%
Perrysville Avenue & Perrysville Road	SSW	1.46	0	1	0.0%
Perry Highway & Good Lane	WNW	1.56	0	0	0.0%
Perrysville Avenue & Cemetery Lane	SSW	1.62	33	63	2.3%
Perry Highway & Lindley Lane	WNW	1.63	0	0	0.0%
Three Degree Road & Babcock Boulevard	NNW	1.68	99	205	7.6%
Peoples Road & Babcock Boulevard	SSE	1.76	33	49	1.8%
Perrysville Avenue & West View Avenue	SSW	1.78	33	63	2.3%
McKnight Road & McIntyre Road	NNW	1.78	37	93	3.5%
Perry Highway & Graham Road	NNW	1.81	71	135	5.0%
Sewickley Oakmont Road & Lombardy Drive	NNW	1.85	71	135	5.0%
Perrysville Avenue & Nelson Run Road	SSW	1.89	33	63	2.3%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Babcock Boulevard & Browns Lane	NNW	1.90	28	70	2.6%
Babcock Boulevard & Roderick Drive	NNW	1.95	28	70	2.6%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

7.0 FUTURE CONDITIONS — NO-BUILD

The No-Build scenario represents future traffic volumes in the 2028 without the proposed development. Background traffic growth is applied at 1.50% per year (compound) over 2 years to current counts. Programmed improvements from the PennDOT Statewide Transportation Improvement Program (STIP) and local capital programs are incorporated into the No-Build network where construction is funded and committed within the analysis horizon.

No-Build horizon year	2028
Growth applied	1.50%/yr × 2 yr
Programmed projects	Per STIP and local CIP — confirm at methodology meeting

8.0 FUTURE CONDITIONS — BUILD

The Build scenario adds distributed project-generated trips to the No-Build network for the 2028 analysis horizon. All 36 study-area intersections and roadway segments are evaluated for LOS under Build conditions.

8.1 Intersection Level of Service — Three-Scenario Summary

Intersection	Ctrl	Dist (mi)	Trips	Exist LOS	No-Bld LOS	Build LOS	Δ delay (s)	Q95 (ft)
Near McKnight Road	Sig	0.38	130	C	C	D	14.2	27
Babcock Boulevard & Rochester Road	Sig	0.40	112	B	B	B	1.4	19
Siebert Road & McKnight Road	Sig	0.41	232	B	B	B	2.0	15
McKnight Road & Braunlich Drive	Sig	0.44	94	C	C	D	6.9	30
Babcock Boulevard & Siebert Road	Sig	0.44	205	B	B	B	3.1	25
McKnight Road & Hilands Place	Sig	0.58	47	C	C	C	1.9	28
Cemetery Lane & Babcock Boulevard	Sig	0.66	112	D	D	E	17.3	20
Glenmore Avenue & Rochester Road	Sig	0.71	78	B	B	B	0.6	17
Babcock Boulevard & Evergreen Road	Sig	0.85	12	B	B	B	0.1	5
McKnight Road & Patrick Place	Sig	0.87	93	C	D	D	11.6	34
Perry Highway & West View Park Drive	Sig	1.07	4	B	B	B	0.0	14
Center Avenue & Perry Highway	Sig	1.08	4	B	B	B	0.0	15
Perry Highway & Chalfonte Avenue	Sig	1.12	3	B	B	B	0.1	14
Perry Highway & Highland Avenue	Sig	1.14	3	B	B	B	0.1	8
McKnight Road & Ross Park Mall Drive	Sig	1.17	93	C	C	D	8.6	37
Bellevue Road & Perry Highway	Sig	1.18	2	B	B	B	0.0	13
Perry Highway & Rochester Road	Sig	1.19	78	B	B	B	0.4	6

Intersection	Ctrl	Dist (mi)	Trips	Exist LOS	No-Bld LOS	Build LOS	Δ delay (s)	Q95 (ft)
McKnight Road & McKnight Circle	Sig	1.19	151	C	C	D	14.9	30
Perry Highway & Perry Lane	Sig	1.21	2	B	B	B	0.0	12
Center Avenue & West View Park Drive	Sig	1.26	2	B	B	B	0.0	4
McKnight Road & Nelson Run Road	Sig	1.29	140	A	A	A	0.5	8
Babcock Boulevard & Thompson Run Road	Sig	1.39	49	B	B	C	1.2	22
McKnight Road & Johnanna Drive	Sig	1.45	93	D	D	E	15.0	34
Perrysville Avenue & Perrysville Road	Sig	1.46	1	B	B	B	0.0	13
Perry Highway & Good Lane	Sig	1.56	0	B	B	B	0.0	13
Perrysville Avenue & Cemetery Lane	Sig	1.62	63	D	D	D	8.2	27
Perry Highway & Lindley Lane	Sig	1.63	0	B	B	B	0.0	12
Three Degree Road & Babcock Boulevard	Sig	1.68	205	C	C	D	12.8	35
Peoples Road & Babcock Boulevard	Sig	1.76	49	B	B	B	0.8	19
Perrysville Avenue & West View Avenue	Sig	1.78	63	F	F	F	15.2	40
McKnight Road & McIntyre Road	Sig	1.78	93	D	D	E	12.9	39
Perry Highway & Graham Road	Sig	1.81	135	B	B	B	0.8	8
Sewickley Oakmont Road & Lombardy Drive	Sig	1.85	135	A	A	B	0.7	8
Perrysville Avenue & Nelson Run Road	Sig	1.89	63	A	A	A	0.3	7
Babcock Boulevard & Browns Lane	Sig	1.90	70	C	D	D	7.8	28
Babcock Boulevard & Roderick Drive	Sig	1.95	70	D	D	D	9.6	28

Acceptable PennDOT LOS threshold: LOS D (urban) / LOS C (rural). Locations shown at Build LOS E or F require mitigation analysis in §9.

9.0 MITIGATION ANALYSIS

The following mitigation measures are proposed to address Build-scenario LOS deficiencies identified in §8. Mitigation must be incorporated as permit conditions in the PennDOT access permit and local site-plan approval.

9.1 Signal Timing Optimization

Signal retiming at affected intersections is proposed to improve cycle lengths and phase splits to accommodate the added project volume. Synchro timing plans should be submitted to the applicable signal operations authority (PennDOT or local traffic engineering) for review and implementation.

9.2 Geometric Improvements

Where signal retiming alone is insufficient to achieve acceptable LOS, geometric improvements (auxiliary turn lanes, channelization) may be required. Turn-lane warrants should be confirmed per the applicable design manual.

9.3 Pedestrian and Multimodal Improvements

Pedestrian crossing enhancements (high-emphasis crosswalks, ADA-compliant ramps per PROWAG) at signalized intersections along the project frontage roads should be coordinated with the applicable agency traffic engineering division.

9.4 Storage Bay Adequacy

95th-percentile queue analysis was performed at driveways and adjacent signalized intersections. Storage bay adequacy was evaluated against projected Build-scenario left-turn and right-turn queue lengths. Where storage deficiencies exist (95th-percentile queue exceeds available storage), turn-lane extensions or new auxiliary lanes are identified as required mitigation.

No intersection carries a field-measured `existingStorageFt` value. Supply that field on each affected intersection record to produce a Required-vs-Existing-vs-Deficit table per AASHTO Green Book Chapter 9 /

applicable state DOT design manual storage-bay standards.

9.5 State-Specific Requirements

- Highway Occupancy Permit (HOP) — PennDOT Form M-945A — required for any access to a state highway.
- PennDOT Pub 282 specifies mandatory report format, contents checklist, and LOS analysis procedures — follow the Pub 282 outline for PennDOT submittals.
- Sight-distance analysis per PennDOT Pub 70 (Engineering and Traffic Studies) or AASHTO Green Book required at all driveways.

10.0 SITE ACCESS AND INTERNAL CIRCULATION

Driveway access geometry, sight distance, and on-site circulation are reviewed per PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual and the applicable access management standards. Access permit requirements: PennDOT Pub 282 defines three categories: Category 1 (≥100 PHT) requires full TIS; Category 2 (50-99 PHT) requires abbreviated TIS; Category 3 (<50 PHT) exempt from TIS but may require an operational analysis. Highway Occupancy Permit (HOP) required for any access to a state highway. The analysis horizon for access-permit purposes is Existing + opening year + 10-year design horizon per PennDOT Pub 282. A secondary 20-year horizon may be required for principal arterials..

Study horizons	Existing + opening year + 10-year design horizon per PennDOT Pub 282. A secondary 20-year horizon may be required for principal arterials.
Access permit authority	PennDOT
Sight distance standard	AASHTO Green Book / state design manual, per design speed

Site access geometry, driveways, internal circulation, and parking design are dependent on the final site plan. A detailed site-access analysis per the applicable design manual should be performed once the site plan is finalized.

11.0 TRANSIT AND MULTIMODAL CONSIDERATIONS

Existing transit service, bicycle facilities, and pedestrian infrastructure within 2.00 miles of the site were inventoried. Any transit-mode reduction applied to trip generation reflects the transit availability factor approved at the methodology meeting. Multimodal improvements (transit shelter upgrades, bicycle parking, pedestrian connections) should be coordinated with the applicable transit provider and included in the site-plan submittal.

Transit reduction applied	None
Nearest transit stop	Verify field
Bicycle facility	Verify field inventory

12.0 CONCLUSIONS AND RECOMMENDATIONS

The proposed McKnight Road Retail Development is projected to generate 1,600 net new external PM peak-hour trips. 10 intersections drop one or more LOS grades and 4 locations operate at LOS E or F under Build conditions. The mitigation measures identified in §9 are required as conditions of the PennDOT access permit to maintain acceptable operations.

Professional Engineer Certification

This Traffic Impact Study (TIS) has been prepared and reviewed by a Professional Engineer licensed in the State of Pennsylvania under Pennsylvania Engineering Practice Act (63 Pa. C.S. § 21 et seq. (Engineering Practice Act); 49 Pa. Code Ch. 37). The signing PE attests to the accuracy and completeness of this study to the degree specified by the controlling PennDOT guidelines.

PE Seal — Pennsylvania
Pennsylvania Engineering Practice Act
63 Pa. C.S. § 21 et seq. (Engineering Practice Act);
49 Pa. Code Ch. 37

Signature: _____
Date: _____

Appendices (referenced — not auto-generated)

- Appendix A — Methodology Meeting Notes / Scope Letter
- Appendix B — Trip Generation Worksheets (public-data rates)
- Appendix C — Traffic Count Data
- Appendix D — Intersection Capacity Analysis Output (HCS / Synchro)
- Appendix E — Signal Warrant Analyses (where applicable)
- Appendix F — Crash Data (3-year, where required by reviewing agency)

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	200 1,000 sqft GLA
Daily trips (productions)	16,000
PM peak-hour trips	1,600 (768 in / 832 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{-(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b=-0.02$, $c=-0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Siebert Road & McKnight Road	0.41	1.0	232	8.6%
Babcock Boulevard & Siebert Road	0.44	1.1	205	7.6%
Three Degree Road & Babcock Boulevard	1.68	4.0	205	7.6%
McKnight Road & McKnight Circle	1.19	2.9	151	5.6%
McKnight Road & Nelson Run Road	1.29	3.1	140	5.2%
Perry Highway & Graham Road	1.81	4.3	135	5.0%
Sewickley Oakmont Road & Lombardy Drive	1.85	4.4	135	5.0%
Near McKnight Road	0.38	0.9	130	4.8%
Babcock Boulevard & Rochester Road	0.40	1.0	112	4.2%
Cemetery Lane & Babcock Boulevard	0.66	1.6	112	4.2%
McKnight Road & Braunlich Drive	0.44	1.1	94	3.5%
McKnight Road & Patrick Place	0.87	2.1	93	3.5%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 77% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's auto-mode share — measured (ACS B08301) where the metro is wired, otherwise the ACS B08301 metropolitan median of 90% — and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 23% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c : 0.60. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	5.52	860	0.60
Minor arterial	3.99	312	0.31
Trunk highway	3.43	173	0.20
Principal arterial	3.50	53	0.16

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 10 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 1,134 network junctions, max imbalance 0.00 veh. 22 study intersection(s) carry path-derived movements; 14 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Study scope: 36 signalized intersections lie within the 2-mile study area, and all 36 are carried as study intersections. The regional signal inventory holds 72 records inside that radius; 36 were merged into a junction already counted, because a signalized junction is commonly recorded once per approach rather than once per intersection. Merged records are duplicates of an analyzed intersection, not intersections omitted from the study.

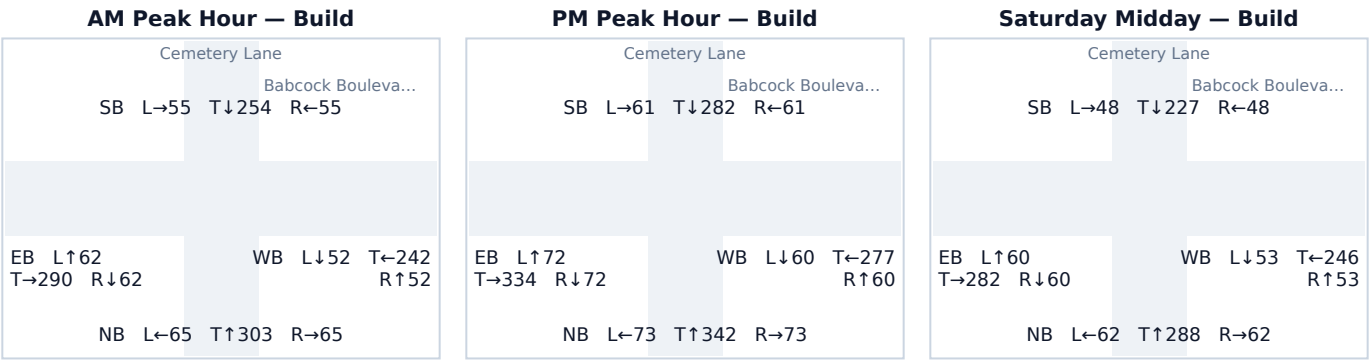
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Cemetery Lane & Babcock Boulevard

Signal ID	pittsburgh-293
Location	NW Pittsburgh · 0.66 mi from site
Added PM peak trips	112
Existing / No-Build (PM)	LOS D · 45.0 s/veh · v/c 0.98
Build (PM)	LOS E · 62.3 s/veh · v/c 1.05
Δ control delay	+17.3 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +17.2s, design year +28.6s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.42 / EW 0.41; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	458	30	488	0.60 → 0.64	16.9 → 17.9	B → B	266
SB	404	0	404	0.53 → 0.53	15.5 → 15.5	B → B	207
EB	445	33	478	0.60 → 0.65	17.5 → 18.5	B → B	264
WB	348	49	397	0.47 → 0.54	15.1 → 16.1	B → B	206

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	49	44%
EB Right	33	29%
NB Left	30	27%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

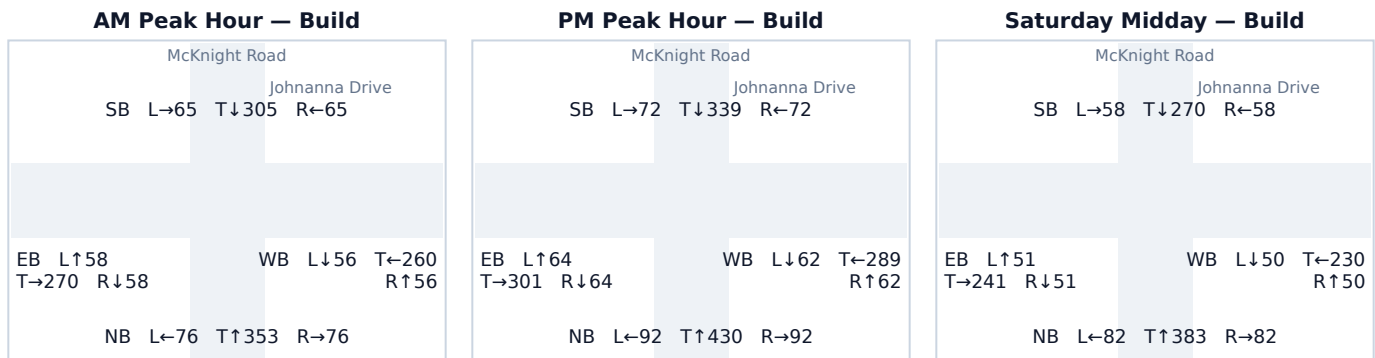
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 McKnight Road & Johnanna Drive

Signal ID	pittsburgh-2913
Location	NW Pittsburgh · 1.45 mi from site
Added PM peak trips	93
Existing / No-Build (PM)	LOS D · 50.0 s/veh · v/c 1.01
Build (PM)	LOS E · 65.0 s/veh · v/c 1.06
Δ control delay	+15.0 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +15.0s, design year +22.4s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.46 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	520	93	614	0.63 → 0.75	16.1 → 19.6	B → B	348
SB	483	0	483	0.59 → 0.59	15.2 → 15.2	B → B	246
EB	429	0	429	0.63 → 0.63	19.8 → 19.8	B → B	241
WB	413	0	413	0.61 → 0.61	19.2 → 19.2	B → B	230

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	93	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

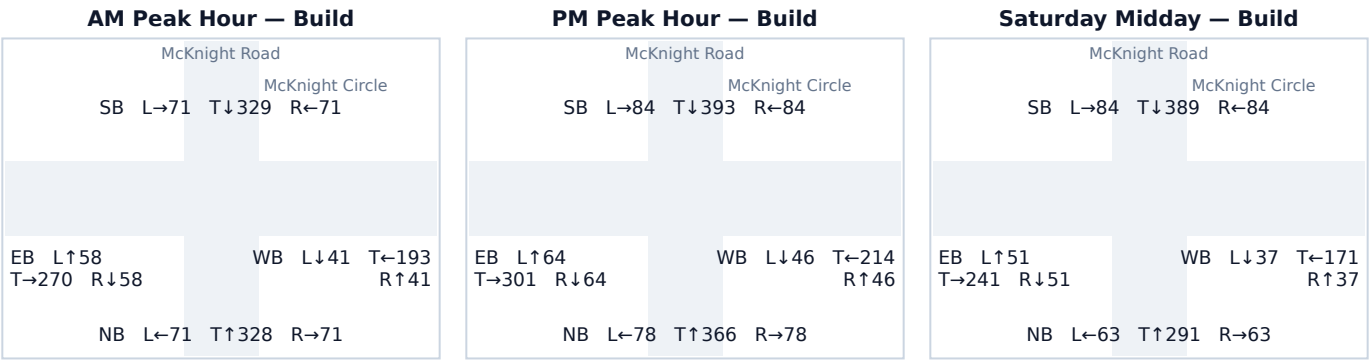
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 McKnight Road & McKnight Circle

Signal ID	pittsburgh-2402
Location	NW Pittsburgh · 1.19 mi from site
Added PM peak trips	151
Existing / No-Build (PM)	LOS C · 31.2 s/veh · v/c 0.91
Build (PM)	LOS D · 46.1 s/veh · v/c 0.99
Δ control delay	+14.9 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +14.9s, design year +35.2s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.46 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	522	0	522	0.63 → 0.63	16.1 → 16.1	B → B	274
SB	410	151	561	0.50 → 0.68	13.6 → 17.4	B → B	304
EB	429	0	429	0.63 → 0.63	19.8 → 19.8	B → B	242
WB	306	0	306	0.45 → 0.45	16.2 → 16.2	B → B	158

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	151	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Near McKnight Road

Signal ID	pittsburgh-2386
Location	NW Pittsburgh · 0.38 mi from site
Added PM peak trips	130
Existing / No-Build (PM)	LOS C · 33.9 s/veh · v/c 0.93
Build (PM)	LOS D · 48.1 s/veh · v/c 1.00
Δ control delay	+14.2 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +14.2s, design year +30.5s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.46 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)

AM Peak Hour — Build	PM Peak Hour — Build	Saturday Midday — Build
Near McKnight Road SB L→67 T↓314 R←67 EB L↑59 T→273 R↓59 WB L↓48 T←227 R↑48 NB L←67 T↑310 R→67	Near McKnight Road SB L→76 T↓354 R←76 EB L↑71 T→330 R↓71 WB L↓55 T←256 R↑55 NB L←74 T↑346 R→74	Near McKnight Road SB L→64 T↓301 R←64 EB L↑63 T→292 R↓63 WB L↓47 T←219 R↑47 NB L←59 T↑277 R→59

Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	494	0	494	0.60 → 0.60	15.3 → 15.3	B → B	253
SB	472	34	506	0.57 → 0.61	14.8 → 15.7	B → B	262
EB	404	68	472	0.60 → 0.70	19.0 → 21.8	B → C	275
WB	338	28	366	0.50 → 0.54	17.0 → 17.8	B → B	197

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	37	28%
SB Right	34	26%
EB Through	31	24%
WB Through	28	22%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

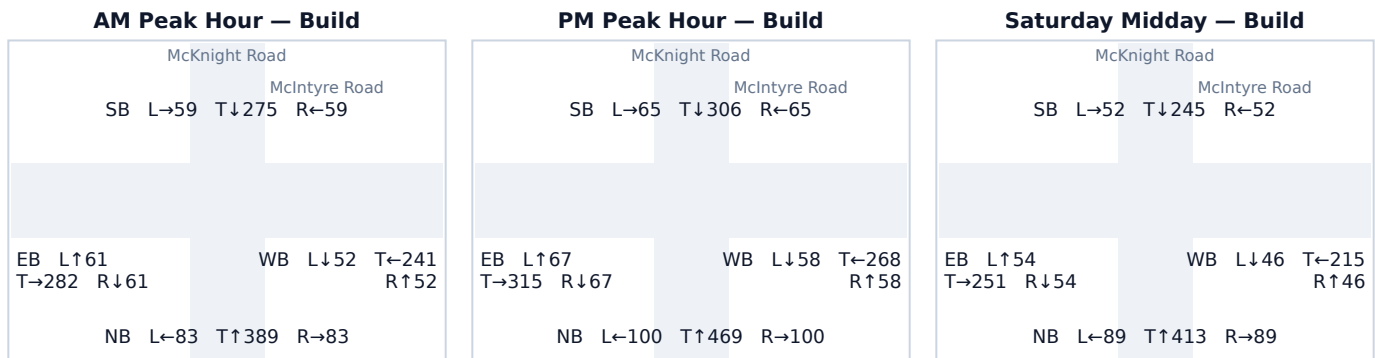
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 McKnight Road & McIntyre Road

Signal ID	pittsburgh-2415
Location	NW Pittsburgh · 1.78 mi from site
Added PM peak trips	93
Existing / No-Build (PM)	LOS D · 43.4 s/veh · v/c 0.99
Build (PM)	LOS E · 56.3 s/veh · v/c 1.04
Δ control delay	+12.9 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +12.9s, design year +21.7s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.47 / EW 0.36; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	576	93	669	0.68 → 0.79	17.0 → 21.1	B → C	390
SB	436	0	436	0.52 → 0.52	13.5 → 13.5	B → B	211
EB	449	0	449	0.68 → 0.68	21.8 → 21.8	C → C	261
WB	384	0	384	0.58 → 0.58	19.1 → 19.1	B → B	213

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	93	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

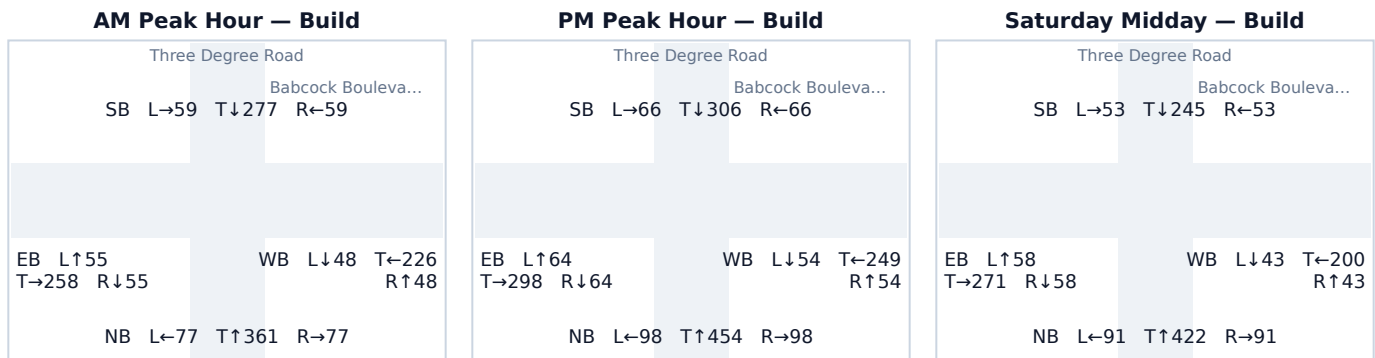
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Three Degree Road & Babcock Boulevard

Signal ID	pittsburgh-285
Location	NW Pittsburgh · 1.68 mi from site
Added PM peak trips	205
Existing / No-Build (PM)	LOS C · 23.8 s/veh · v/c 0.85
Build (PM)	LOS D · 36.6 s/veh · v/c 0.96
Δ control delay	+12.8 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +12.8s, design year +43.4s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.49 / EW 0.34; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	510	140	650	0.58 → 0.74	13.8 → 17.9	B → B	358
SB	438	0	438	0.50 → 0.50	12.4 → 12.4	B → B	204
EB	361	65	426	0.58 → 0.68	20.0 → 22.9	C → C	251
WB	357	0	357	0.58 → 0.58	19.9 → 19.9	B → B	201

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Left	70	34%
NB Right	70	34%
EB Right	65	32%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

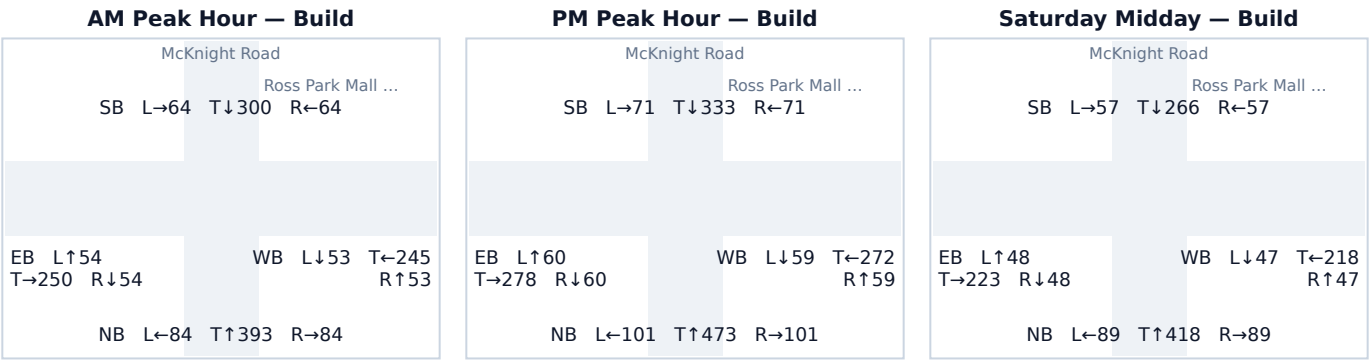
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 McKnight Road & Ross Park Mall Drive

Signal ID	pittsburgh-2404
Location	NW Pittsburgh · 1.17 mi from site
Added PM peak trips	93
Existing / No-Build (PM)	LOS C · 31.9 s/veh · v/c 0.93
Build (PM)	LOS D · 40.5 s/veh · v/c 0.98
Δ control delay	+8.6 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +8.6s, design year +20.3s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.49 / EW 0.34; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	582	93	675	0.65 → 0.76	15.0 → 18.3	B → B	375
SB	475	0	475	0.53 → 0.53	12.7 → 12.7	B → B	224
EB	398	0	398	0.65 → 0.65	22.2 → 22.2	C → C	232
WB	390	0	390	0.64 → 0.64	21.8 → 21.8	C → C	226

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	93	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

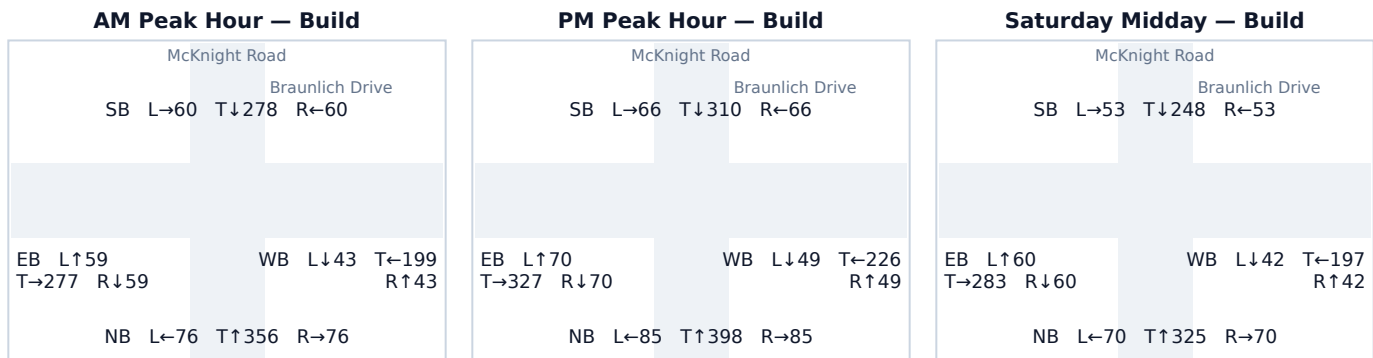
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 McKnight Road & Braunlich Drive

Signal ID	pittsburgh-2393
Location	NW Pittsburgh · 0.44 mi from site
Added PM peak trips	94
Existing / No-Build (PM)	LOS C · 28.7 s/veh · v/c 0.90
Build (PM)	LOS D · 35.6 s/veh · v/c 0.95
Δ control delay	+6.9 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +6.9s, design year +20.8s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.47 / EW 0.36; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	553	14	568	0.65 → 0.66	15.7 → 16.1	B → B	300
SB	442	0	442	0.52 → 0.52	13.2 → 13.2	B → B	212
EB	418	49	467	0.65 → 0.72	21.0 → 23.6	C → C	278
WB	294	31	324	0.46 → 0.50	17.1 → 17.8	B → B	174

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	33	35%
WB Through	31	33%
EB Right	16	17%
NB Left	14	15%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

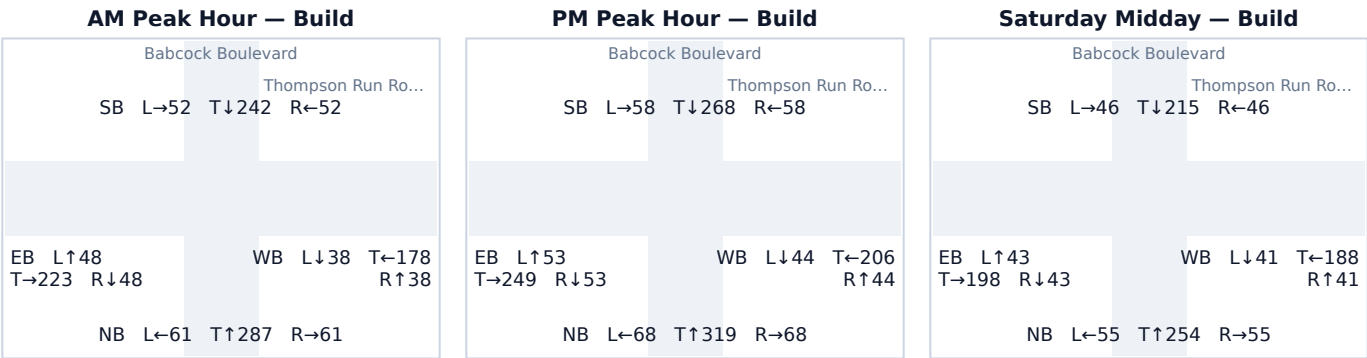
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Babcock Boulevard & Thompson Run Road

Signal ID	pittsburgh-292
Location	NE Pittsburgh · 1.39 mi from site
Added PM peak trips	49
Existing / No-Build (PM)	LOS B · 19.9 s/veh · v/c 0.77
Build (PM)	LOS C · 21.1 s/veh · v/c 0.79
Δ control delay	+1.2 s/veh (LOS change)
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback. Governing horizon: design year (opening year +1.2s, design year +8.1s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.47 / EW 0.36; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	455	0	455	0.54 → 0.54	13.8 → 13.8	B → B	222
SB	384	0	384	0.46 → 0.46	12.6 → 12.6	B → B	179
EB	355	0	355	0.54 → 0.54	18.2 → 18.2	B → B	193
WB	246	49	294	0.37 → 0.45	15.6 → 16.6	B → B	154

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	49	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

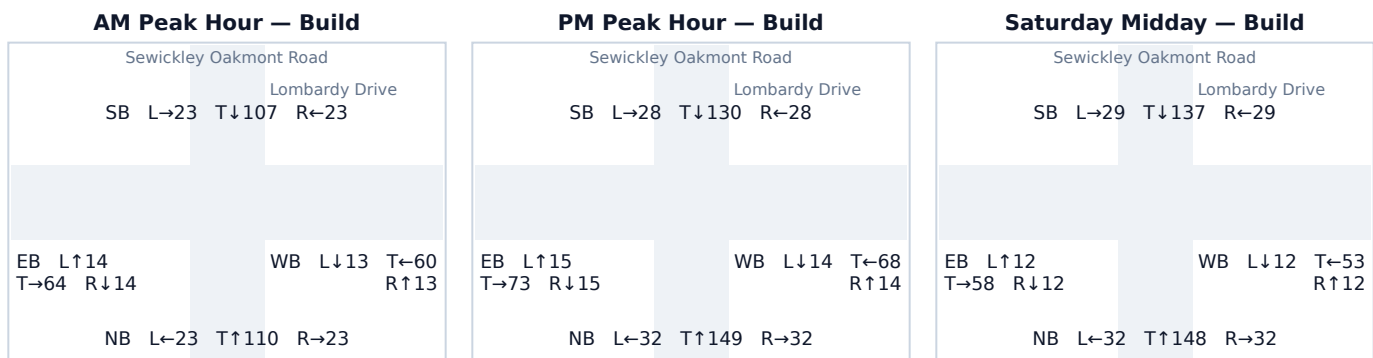
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Sewickley Oakmont Road & Lombardy Drive

Signal ID	pittsburgh-2220
Location	NW Pittsburgh · 1.85 mi from site
Added PM peak trips	135
Existing / No-Build (PM)	LOS A · 9.6 s/veh · v/c 0.24
Build (PM)	LOS B · 10.3 s/veh · v/c 0.31
Δ control delay	+0.7 s/veh (LOS change)
Mitigation	No mitigation indicated at screening level (opening year +0.6s, design year +0.7s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.48 / EW 0.35; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	143	70	213	0.16 → 0.24	9.0 → 9.7	A → A	85
SB	121	65	186	0.14 → 0.21	8.9 → 9.4	A → A	74
EB	103	0	103	0.16 → 0.16	14.1 → 14.1	B → B	49
WB	96	0	96	0.15 → 0.15	14.0 → 14.0	B → B	45

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	70	52%
SB Through	65	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

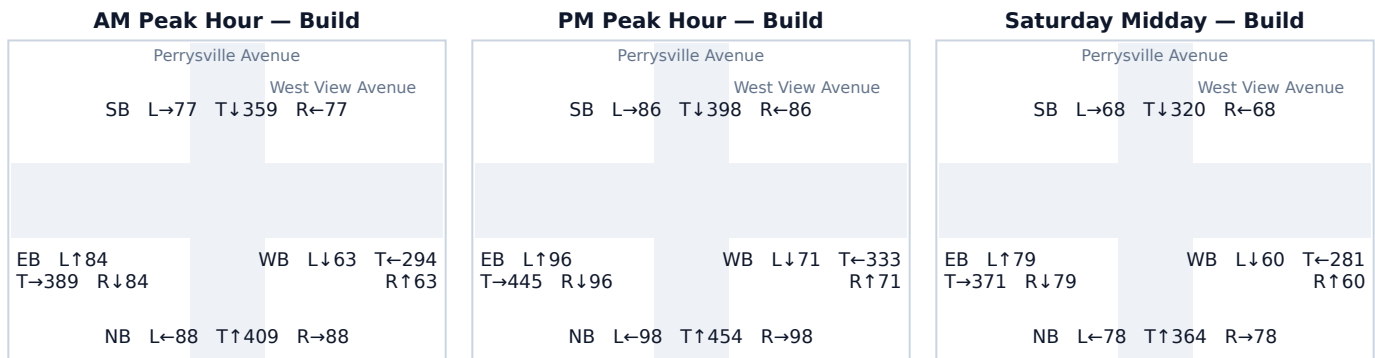
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Perrysville Avenue & West View Avenue

Signal ID	pittsburgh-2208
Location	Central Pittsburgh · 1.78 mi from site
Added PM peak trips	63
Existing / No-Build (PM)	LOS F · 153.7 s/veh · v/c 1.28
Build (PM)	LOS F · 168.9 s/veh · v/c 1.31
Δ control delay	+15.2 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +15.2s, design year +0.0s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 70 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.41; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	650	0	650	0.81 → 0.81	25.8 → 25.8	C → C	453
SB	570	0	570	0.71 → 0.71	21.2 → 21.2	C → C	372
EB	604	33	637	0.81 → 0.86	27.6 → 30.8	C → C	464
WB	444	30	475	0.60 → 0.64	19.5 → 20.5	B → C	303

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	33	52%
WB Through	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

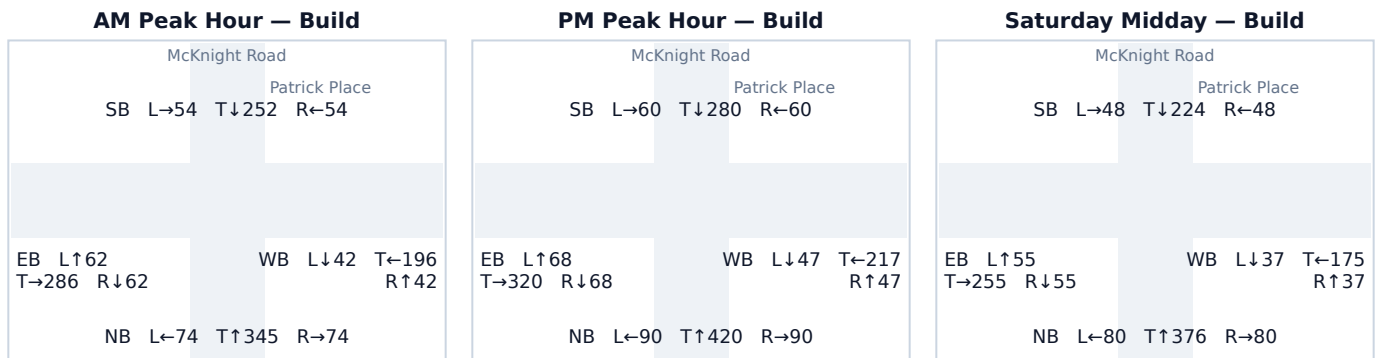
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 McKnight Road & Patrick Place

Signal ID	pittsburgh-2407
Location	NW Pittsburgh · 0.87 mi from site
Added PM peak trips	93
Existing / No-Build (PM)	LOS D · 38.8 s/veh · v/c 0.95
Build (PM)	LOS D · 50.4 s/veh · v/c 1.01
Δ control delay	+11.6 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +11.7s, design year +22.9s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.40; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	507	93	600	0.64 → 0.76	17.1 → 21.0	B → C	347
SB	400	0	400	0.51 → 0.51	14.5 → 14.5	B → B	199
EB	456	0	456	0.64 → 0.64	19.1 → 19.1	B → B	254
WB	311	0	311	0.44 → 0.44	15.2 → 15.2	B → B	156

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	93	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

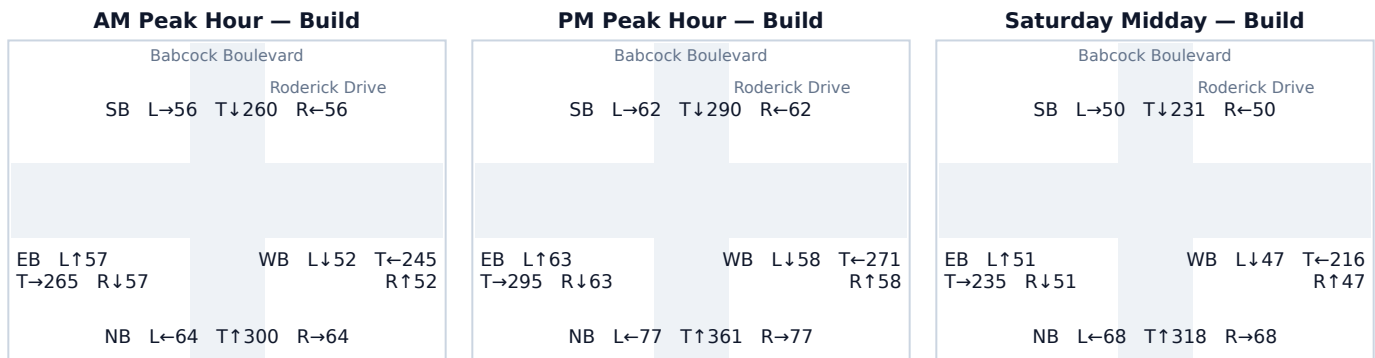
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.13 Babcock Boulevard & Roderick Drive

Signal ID	pittsburgh-286
Location	NW Pittsburgh · 1.95 mi from site
Added PM peak trips	70
Existing / No-Build (PM)	LOS D · 43.4 s/veh · v/c 0.97
Build (PM)	LOS D · 53.0 s/veh · v/c 1.01
Δ control delay	+9.6 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +9.7s, design year +17.7s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.43 / EW 0.41; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	445	70	515	0.58 → 0.67	16.2 → 18.3	B → B	284
SB	414	0	414	0.54 → 0.54	15.4 → 15.4	B → B	211
EB	421	0	421	0.58 → 0.58	17.2 → 17.2	B → B	225
WB	387	0	387	0.53 → 0.53	16.3 → 16.3	B → B	202

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	70	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

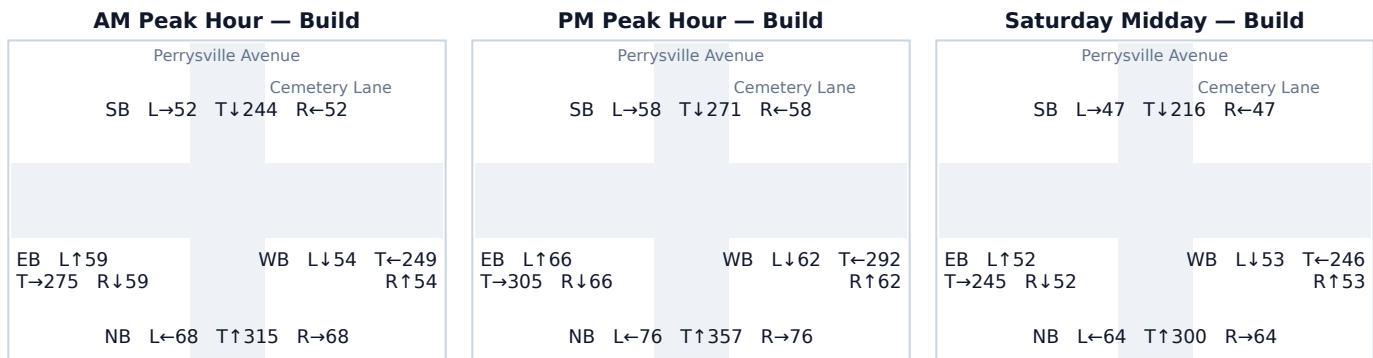
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.14 Perrysville Avenue & Cemetery Lane

Signal ID	pittsburgh-1826
Location	Central Pittsburgh · 1.62 mi from site
Added PM peak trips	63
Existing / No-Build (PM)	LOS D · 41.4 s/veh · v/c 0.97
Build (PM)	LOS D · 49.6 s/veh · v/c 1.00
Δ control delay	+8.2 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +8.1s, design year +15.6s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.40; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	478	30	509	0.61 → 0.65	16.5 → 17.4	B → B	275
SB	387	0	387	0.49 → 0.49	14.4 → 14.4	B → B	191
EB	437	0	437	0.61 → 0.61	18.2 → 18.2	B → B	239
WB	383	33	416	0.53 → 0.58	16.7 → 17.6	B → B	224

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	33	52%
NB Right	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

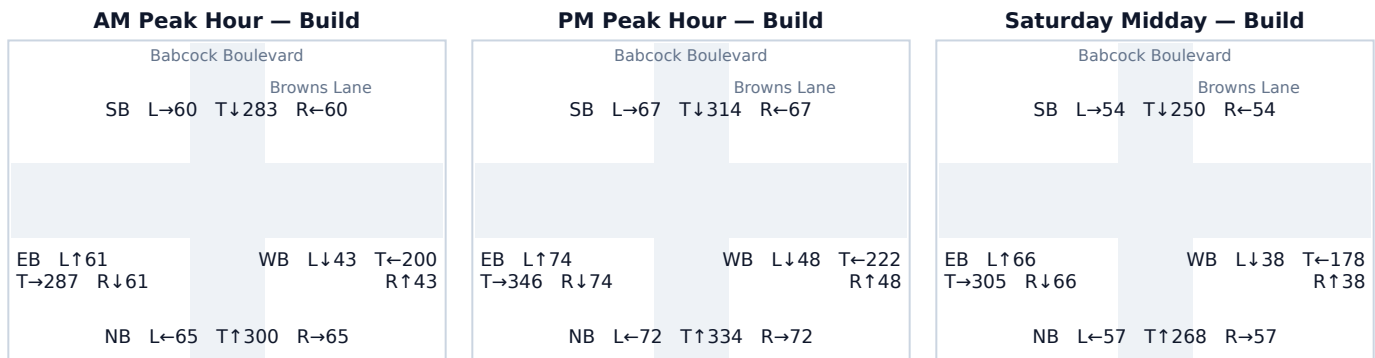
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.15 Babcock Boulevard & Browns Lane

Signal ID	pittsburgh-1393
Location	NW Pittsburgh · 1.90 mi from site
Added PM peak trips	70
Existing / No-Build (PM)	LOS D · 36.9 s/veh · v/c 0.94
Build (PM)	LOS D · 44.7 s/veh · v/c 0.98
Δ control delay	+7.8 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +7.8s, design year +17.0s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.39; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	478	0	478	0.60 → 0.60	16.1 → 16.1	B → B	250
SB	448	0	448	0.56 → 0.56	15.3 → 15.3	B → B	229
EB	424	70	494	0.60 → 0.70	18.3 → 21.0	B → C	285
WB	318	0	318	0.45 → 0.45	15.6 → 15.6	B → B	161

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	70	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

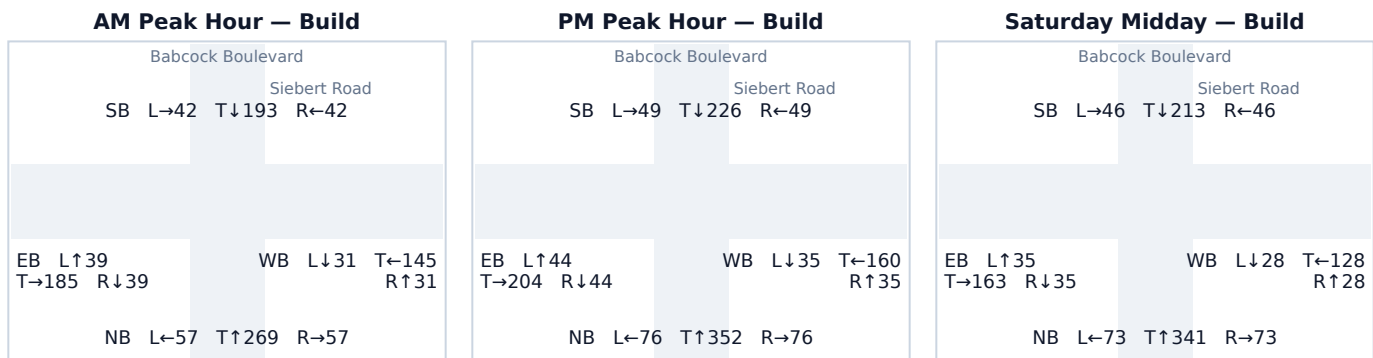
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.16 Babcock Boulevard & Siebert Road

Signal ID	pittsburgh-295
Location	NW Pittsburgh · 0.44 mi from site
Added PM peak trips	205
Existing / No-Build (PM)	LOS B · 15.6 s/veh · v/c 0.62
Build (PM)	LOS B · 18.7 s/veh · v/c 0.73
Δ control delay	+3.1 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +3.1s, design year +12.0s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.46 / EW 0.37; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	364	140	504	0.44 → 0.61	12.5 → 15.3	B → B	259
SB	259	65	324	0.31 → 0.39	11.1 → 11.9	B → B	146
EB	292	0	292	0.44 → 0.44	16.2 → 16.2	B → B	151
WB	230	0	230	0.34 → 0.34	15.0 → 15.0	B → B	114

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	140	68%
SB Through	65	32%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

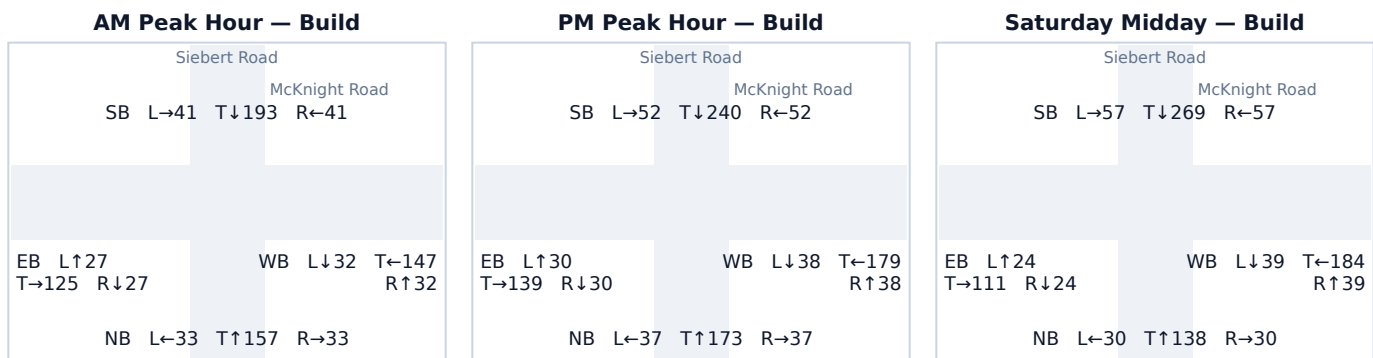
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.17 Siebert Road & McKnight Road

Signal ID	pittsburgh-2382
Location	NW Pittsburgh · 0.41 mi from site
Added PM peak trips	232
Existing / No-Build (PM)	LOS B · 12.6 s/veh · v/c 0.44
Build (PM)	LOS B · 14.6 s/veh · v/c 0.57
Δ control delay	+2.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +1.9s, design year +3.2s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.46 / EW 0.37; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/l_n × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	247	0	247	0.30 → 0.30	11.0 → 11.0	B → B	106
SB	193	151	344	0.23 → 0.41	10.4 → 12.3	B → B	157
EB	199	0	199	0.30 → 0.30	14.5 → 14.5	B → B	97
WB	174	81	255	0.26 → 0.38	14.0 → 15.4	B → B	128

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	151	65%
WB Through	81	35%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

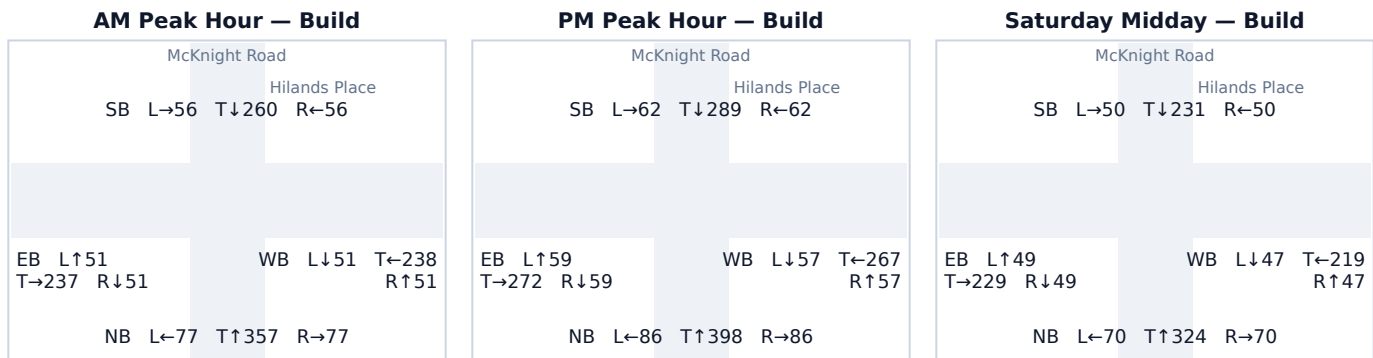
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.18 McKnight Road & Hilands Place

Signal ID	pittsburgh-2397
Location	NW Pittsburgh · 0.58 mi from site
Added PM peak trips	47
Existing / No-Build (PM)	LOS C · 22.8 s/veh · v/c 0.85
Build (PM)	LOS C · 24.7 s/veh · v/c 0.87
Δ control delay	+1.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +1.9s, design year +9.4s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.50 / EW 0.33; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	560	11	570	0.62 → 0.63	14.0 → 14.2	B → B	286
SB	413	0	413	0.46 → 0.46	11.3 → 11.3	B → B	184
EB	365	24	390	0.61 → 0.65	21.5 → 22.7	C → C	229
WB	369	12	381	0.62 → 0.64	21.7 → 22.2	C → C	222

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	13	28%
WB Through	12	26%
EB Right	11	23%
NB Left	11	23%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

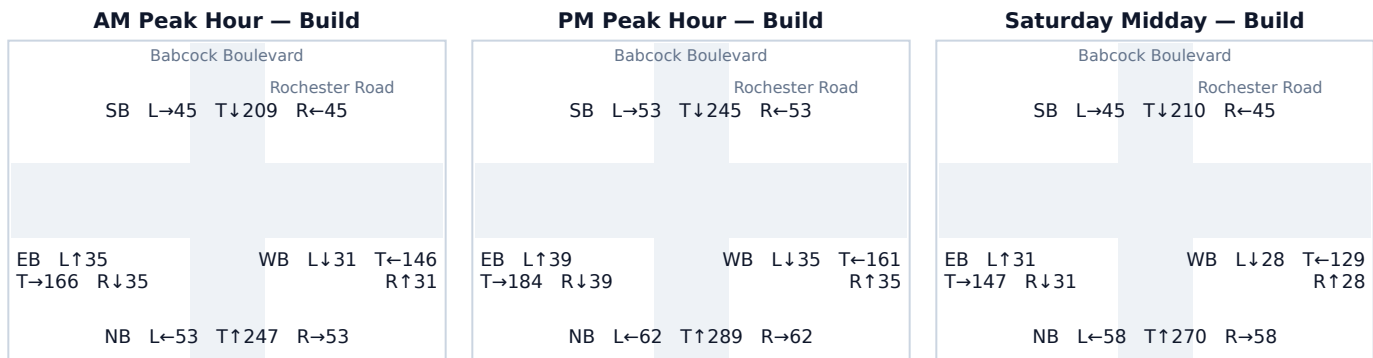
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.19 Babcock Boulevard & Rochester Road

Signal ID	pittsburgh-294
Location	NW Pittsburgh · 0.40 mi from site
Added PM peak trips	112
Existing / No-Build (PM)	LOS B · 15.3 s/veh · v/c 0.61
Build (PM)	LOS B · 16.7 s/veh · v/c 0.67
Δ control delay	+1.4 s/veh
Mitigation	No mitigation indicated at screening level (opening year +1.4s, design year +4.7s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.47 / EW 0.37; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	334	79	413	0.40 → 0.49	11.9 → 13.1	B → B	196
SB	318	33	351	0.38 → 0.42	11.6 → 12.1	B → B	160
EB	262	0	262	0.40 → 0.40	15.9 → 15.9	B → B	134
WB	231	0	231	0.35 → 0.35	15.3 → 15.3	B → B	116

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	79	71%
SB Through	33	29%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

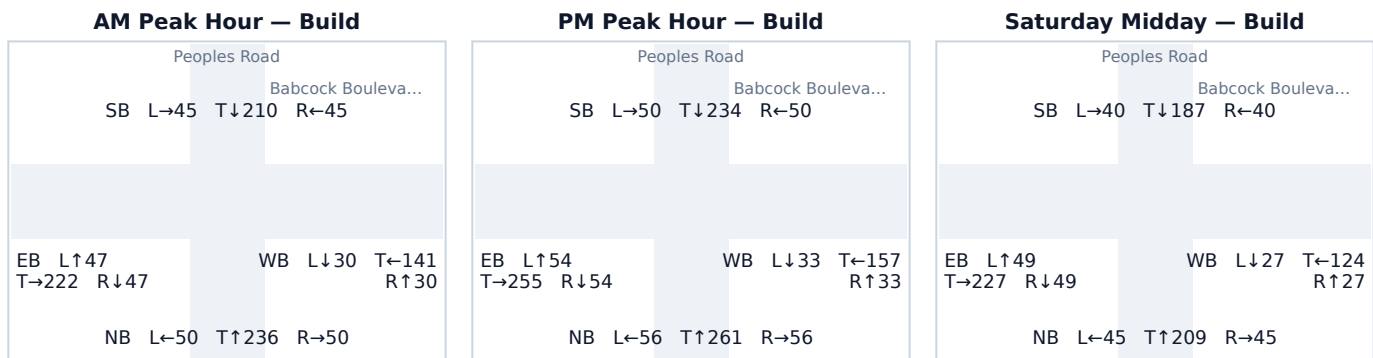
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.20 Peoples Road & Babcock Boulevard

Signal ID	pittsburgh-297
Location	Central Pittsburgh · 1.76 mi from site
Added PM peak trips	49
Existing / No-Build (PM)	LOS B · 17.8 s/veh · v/c 0.69
Build (PM)	LOS B · 18.6 s/veh · v/c 0.71
Δ control delay	+0.8 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +0.8s, design year +4.3s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	373	0	373	0.46 → 0.46	13.2 → 13.2	B → B	177
SB	334	0	334	0.41 → 0.41	12.6 → 12.6	B → B	154
EB	315	49	363	0.46 → 0.53	16.1 → 17.3	B → B	194
WB	223	0	223	0.33 → 0.33	14.4 → 14.4	B → B	108

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	49	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

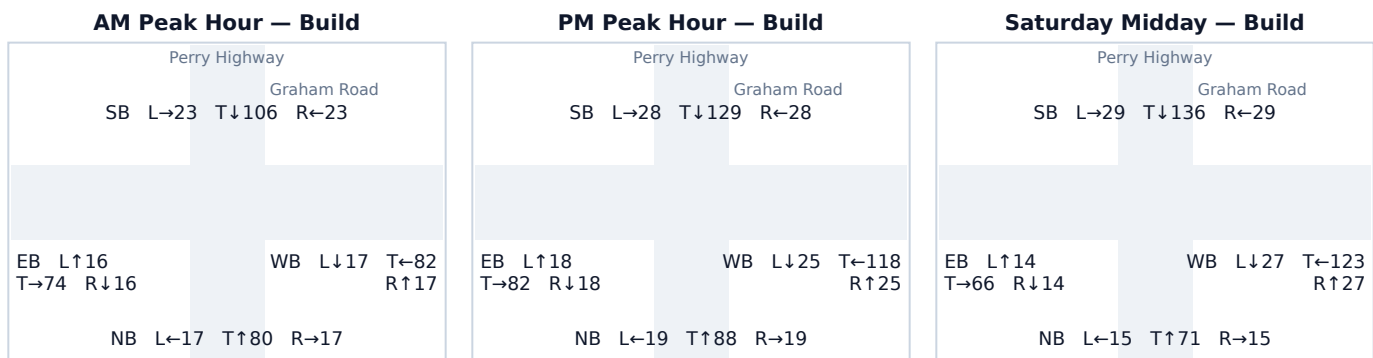
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.21 Perry Highway & Graham Road

Signal ID	pittsburgh-2221
Location	NW Pittsburgh · 1.81 mi from site
Added PM peak trips	135
Existing / No-Build (PM)	LOS B · 11.8 s/veh · v/c 0.27
Build (PM)	LOS B · 12.6 s/veh · v/c 0.35
Δ control delay	+0.8 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.8s, design year +1.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.43 / EW 0.40; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	126	0	126	0.16 → 0.16	10.9 → 10.9	B → B	53
SB	120	65	185	0.16 → 0.24	10.8 → 11.5	B → B	81
EB	118	0	118	0.16 → 0.16	12.0 → 12.0	B → B	52
WB	98	70	168	0.14 → 0.23	11.7 → 12.6	B → B	76

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Right	70	52%
SB Left	65	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

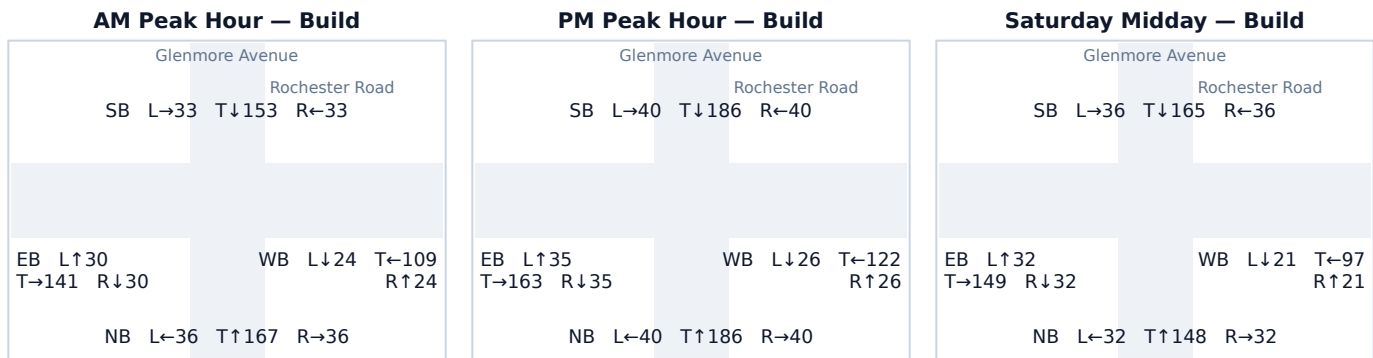
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.22 Glenmore Avenue & Rochester Road

Signal ID	pittsburgh-109
Location	NW Pittsburgh · 0.71 mi from site
Added PM peak trips	78
Existing / No-Build (PM)	LOS B · 12.0 s/veh · v/c 0.45
Build (PM)	LOS B · 12.6 s/veh · v/c 0.49
Δ control delay	+0.6 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.6s, design year +0.9s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.48 / EW 0.35; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	266	0	266	0.31 → 0.31	10.4 → 10.4	B → B	111
SB	225	41	266	0.26 → 0.31	10.0 → 10.4	A → B	111
EB	195	37	233	0.31 → 0.37	15.3 → 16.0	B → B	119
WB	174	0	174	0.27 → 0.27	15.0 → 15.0	B → B	86

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	41	53%
EB Left	37	47%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

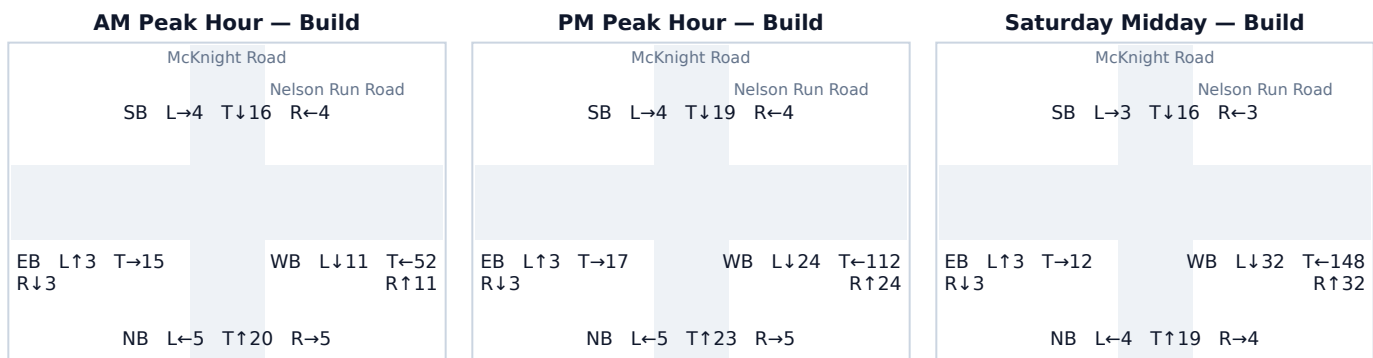
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.23 McKnight Road & Nelson Run Road

Signal ID	pittsburgh-2379
Location	Central Pittsburgh · 1.29 mi from site
Added PM peak trips	140
Existing / No-Build (PM)	LOS A · 8.0 s/veh · v/c 0.05
Build (PM)	LOS A · 8.5 s/veh · v/c 0.12
Δ control delay	+0.5 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.5s, design year +0.5s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.49 / EW 0.34; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/l_n × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	33	0	33	0.04 → 0.04	7.9 → 7.9	A → A	12
SB	27	0	27	0.03 → 0.03	7.9 → 7.9	A → A	10
EB	23	0	23	0.04 → 0.04	13.4 → 13.4	B → B	11
WB	20	140	160	0.03 → 0.26	13.3 → 15.4	B → B	80

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	140	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

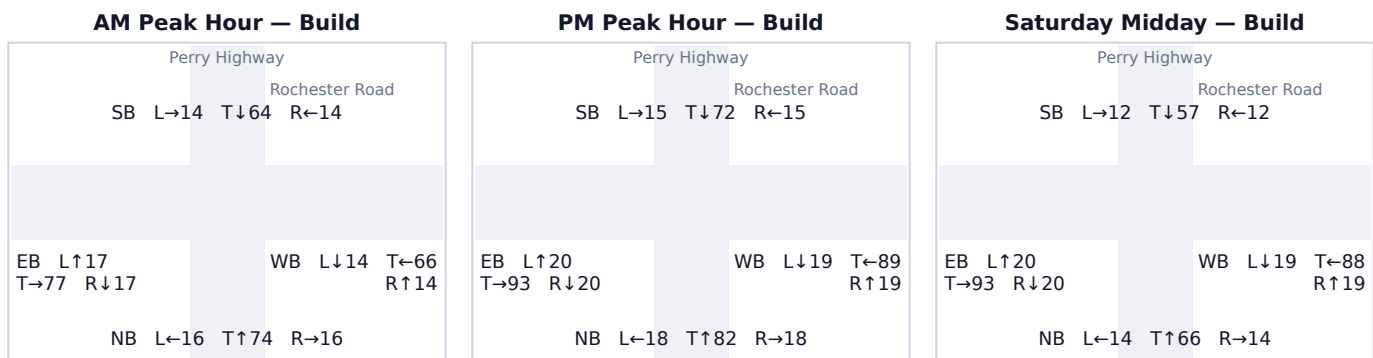
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.24 Perry Highway & Rochester Road

Signal ID	pittsburgh-2228
Location	NW Pittsburgh · 1.19 mi from site
Added PM peak trips	78
Existing / No-Build (PM)	LOS B · 10.3 s/veh · v/c 0.22
Build (PM)	LOS B · 10.7 s/veh · v/c 0.26
Δ control delay	+0.4 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.4s, design year +0.4s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.46 / EW 0.37; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/l_n × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	118	0	118	0.14 → 0.14	9.7 → 9.7	A → A	47
SB	102	0	102	0.12 → 0.12	9.6 → 9.6	A → A	40
EB	96	37	133	0.14 → 0.20	12.9 → 13.4	B → B	62
WB	87	41	127	0.13 → 0.19	12.8 → 13.3	B → B	59

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	41	53%
EB Through	37	47%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

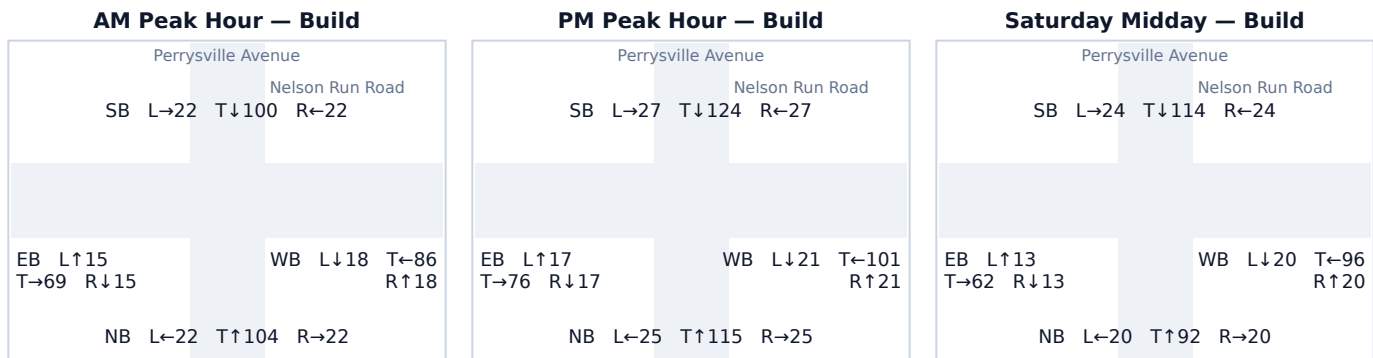
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.25 Perrysville Avenue & Nelson Run Road

Signal ID	pittsburgh-2212
Location	Central Pittsburgh · 1.89 mi from site
Added PM peak trips	63
Existing / No-Build (PM)	LOS A · 9.6 s/veh · v/c 0.27
Build (PM)	LOS A · 9.9 s/veh · v/c 0.30
Δ control delay	+0.3 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.3s, design year +0.3s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.49 / EW 0.34; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	165	0	165	0.19 → 0.19	8.9 → 8.9	A → A	63
SB	145	33	178	0.16 → 0.20	8.7 → 9.0	A → A	69
EB	110	0	110	0.18 → 0.18	14.6 → 14.6	B → B	53
WB	113	30	143	0.19 → 0.23	14.6 → 15.1	B → B	71

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Left	33	52%
WB Right	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

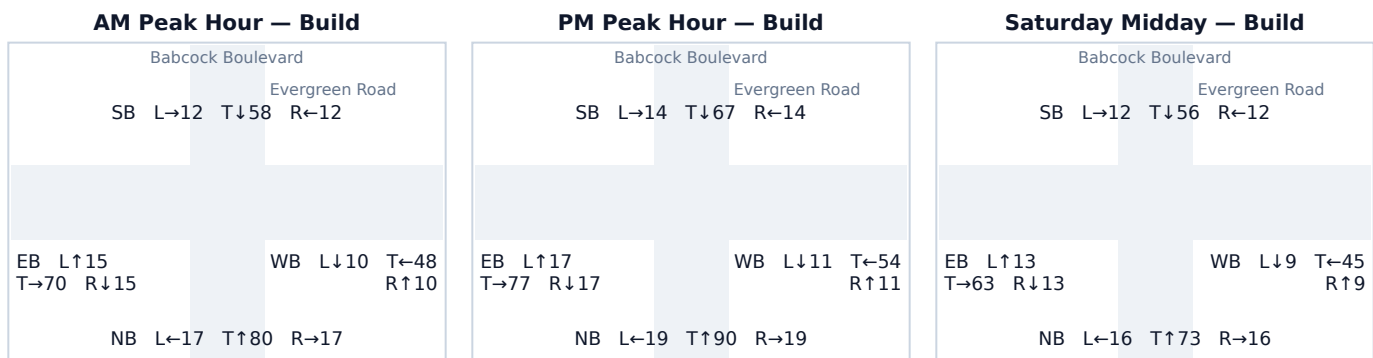
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.26 Babcock Boulevard & Evergreen Road

Signal ID	pittsburgh-2400
Location	NW Pittsburgh · 0.85 mi from site
Added PM peak trips	12
Existing / No-Build (PM)	LOS B · 11.1 s/veh · v/c 0.23
Build (PM)	LOS B · 11.2 s/veh · v/c 0.23
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.1s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.39; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/l_n × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	124	4	128	0.16 → 0.16	10.5 → 10.6	B → B	53
SB	89	6	95	0.11 → 0.12	10.2 → 10.2	B → B	39
EB	111	0	111	0.16 → 0.16	12.2 → 12.2	B → B	49
WB	74	2	76	0.10 → 0.11	11.8 → 11.8	B → B	33

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	4	33%
NB Through	4	33%
SB Left	2	17%
WB Right	2	17%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

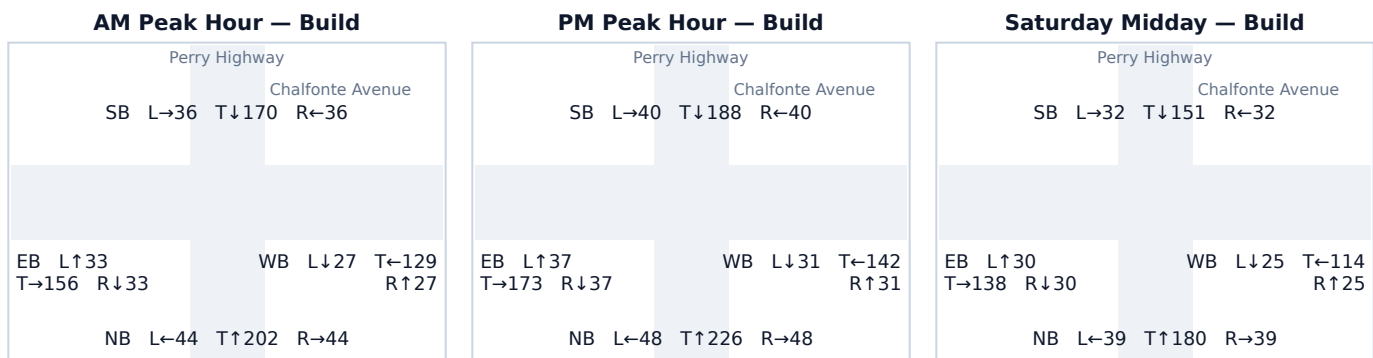
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.27 Perry Highway & Chalfonte Avenue

Signal ID	pittsburgh-813
Location	NW Pittsburgh · 1.12 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS B · 13.8 s/veh · v/c 0.55
Build (PM)	LOS B · 13.9 s/veh · v/c 0.55
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.47 / EW 0.36; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/l_n × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	321	0	322	0.38 → 0.38	11.5 → 11.5	B → B	142
SB	268	0	268	0.32 → 0.32	10.8 → 10.8	B → B	115
EB	246	1	247	0.38 → 0.38	15.9 → 15.9	B → B	126
WB	202	2	204	0.31 → 0.31	15.0 → 15.1	B → B	101

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	33%
EB Through	1	33%
WB Left	1	33%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.28 Perry Highway & Highland Avenue

Signal ID	pittsburgh-2229
Location	NW Pittsburgh · 1.14 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS B · 13.6 s/veh · v/c 0.39
Build (PM)	LOS B · 13.7 s/veh · v/c 0.39
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.42 / EW 0.42; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)

AM Peak Hour — Build	PM Peak Hour — Build	Saturday Midday — Build

Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	186	0	187	0.25 → 0.25	12.2 → 12.2	B → B	84
SB	151	0	151	0.20 → 0.20	11.8 → 11.8	B → B	66
EB	187	1	188	0.25 → 0.25	12.2 → 12.2	B → B	84
WB	119	2	121	0.16 → 0.16	11.4 → 11.4	B → B	52

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	33%
EB Through	1	33%
WB Left	1	33%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

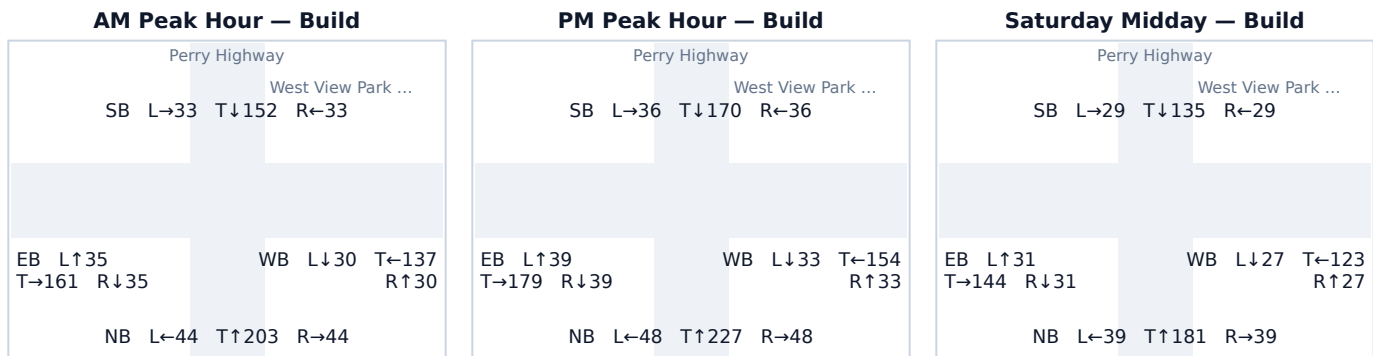
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.29 Perry Highway & West View Park Drive

Signal ID	pittsburgh-665
Location	NW Pittsburgh · 1.07 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 14.3 s/veh · v/c 0.56
Build (PM)	LOS B · 14.3 s/veh · v/c 0.56
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.47 / EW 0.37; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	322	1	323	0.39 → 0.39	11.8 → 11.8	B → B	145
SB	242	0	242	0.29 → 0.29	10.8 → 10.8	B → B	103
EB	255	1	257	0.39 → 0.39	15.6 → 15.7	B → B	130
WB	218	2	220	0.33 → 0.33	14.9 → 15.0	B → B	109

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	25%
EB Through	1	25%
WB Left	1	25%
NB Right	1	25%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

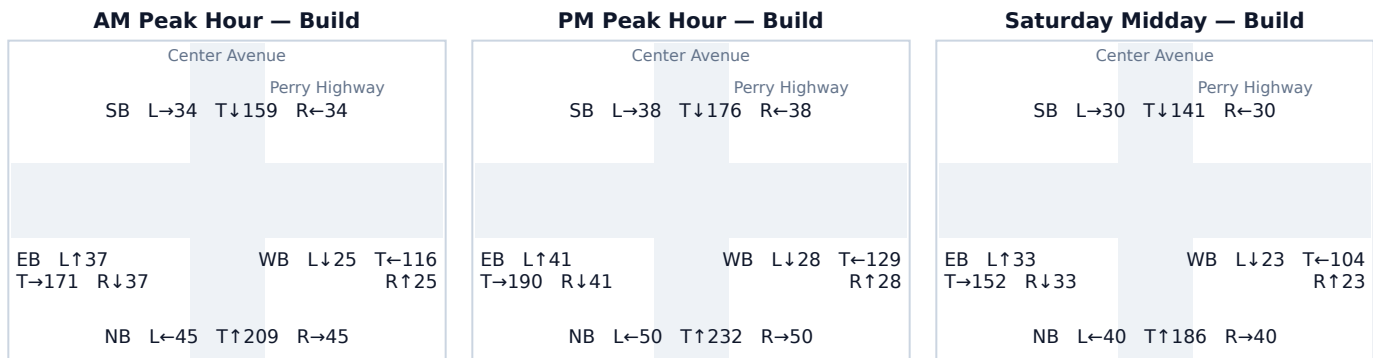
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.30 Center Avenue & Perry Highway

Signal ID	pittsburgh-2217
Location	NW Pittsburgh · 1.08 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 14.7 s/veh · v/c 0.57
Build (PM)	LOS B · 14.7 s/veh · v/c 0.57
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.46 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	332	1	332	0.40 → 0.40	12.2 → 12.2	B → B	152
SB	252	0	252	0.30 → 0.30	11.2 → 11.2	B → B	109
EB	271	1	272	0.40 → 0.40	15.6 → 15.6	B → B	138
WB	183	2	185	0.27 → 0.27	14.1 → 14.1	B → B	89

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	25%
EB Through	1	25%
WB Left	1	25%
NB Right	1	25%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

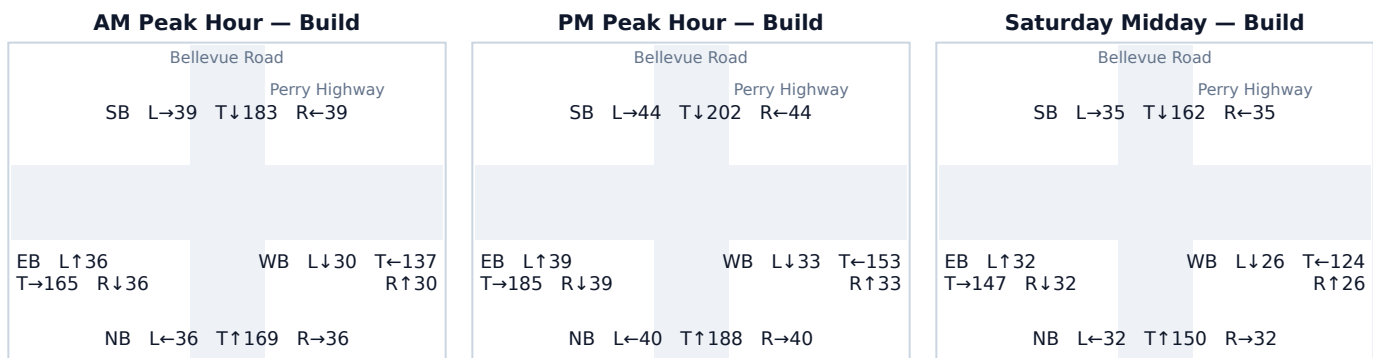
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.31 Bellevue Road & Perry Highway

Signal ID	pittsburgh-1260
Location	NW Pittsburgh · 1.18 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS B · 16.1 s/veh · v/c 0.59
Build (PM)	LOS B · 16.1 s/veh · v/c 0.59
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.40; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	267	0	268	0.34 → 0.34	12.3 → 12.3	B → B	122
SB	290	0	290	0.37 → 0.37	12.6 → 12.6	B → B	134
EB	262	1	263	0.37 → 0.37	14.3 → 14.3	B → B	128
WB	218	1	219	0.31 → 0.31	13.6 → 13.6	B → B	104

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	50%
EB Through	1	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

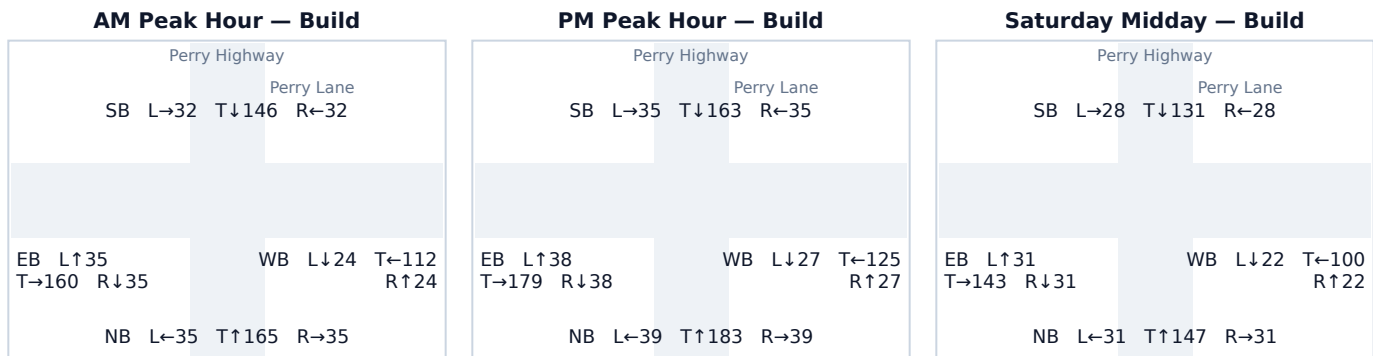
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.32 Perry Highway & Perry Lane

Signal ID	pittsburgh-2227
Location	NW Pittsburgh · 1.21 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS B · 15.9 s/veh · v/c 0.55
Build (PM)	LOS B · 15.9 s/veh · v/c 0.55
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.42 / EW 0.41; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	261	0	261	0.34 → 0.34	13.0 → 13.0	B → B	121
SB	233	0	233	0.31 → 0.31	12.6 → 12.6	B → B	107
EB	255	1	255	0.34 → 0.34	13.4 → 13.4	B → B	120
WB	178	1	179	0.24 → 0.24	12.3 → 12.3	B → B	80

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	50%
EB Through	1	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

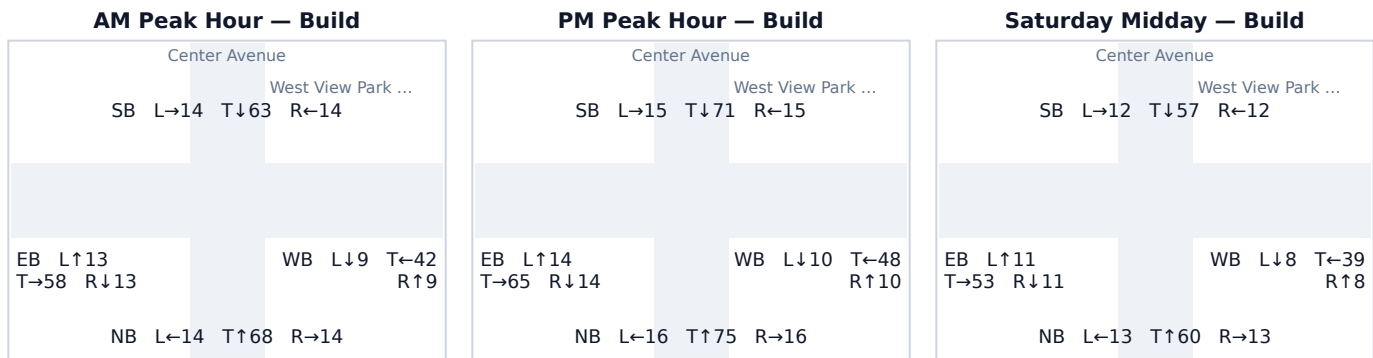
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.33 Center Avenue & West View Park Drive

Signal ID	pittsburgh-728
Location	NW Pittsburgh · 1.26 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS B · 10.7 s/veh · v/c 0.21
Build (PM)	LOS B · 10.7 s/veh · v/c 0.21
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.39; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/l_n × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	107	0	107	0.13 → 0.13	10.1 → 10.1	B → B	43
SB	101	0	101	0.13 → 0.13	10.1 → 10.1	B → B	41
EB	93	1	93	0.13 → 0.13	12.3 → 12.3	B → B	41
WB	67	1	68	0.10 → 0.10	12.0 → 12.0	B → B	30

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	50%
EB Through	1	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

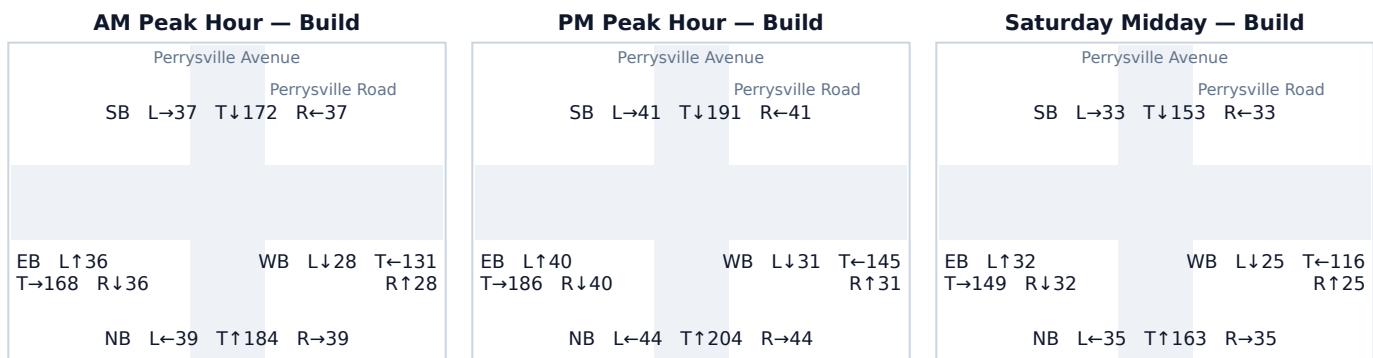
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.34 Perrysville Avenue & Perrysville Road

Signal ID	pittsburgh-2219
Location	NW Pittsburgh · 1.46 mi from site
Added PM peak trips	1
Existing / No-Build (PM)	LOS B · 16.2 s/veh · v/c 0.60
Build (PM)	LOS B · 16.2 s/veh · v/c 0.60
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.40; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	292	0	292	0.37 → 0.37	12.8 → 12.8	B → B	135
SB	273	1	273	0.35 → 0.35	12.5 → 12.5	B → B	125
EB	266	0	266	0.37 → 0.37	14.2 → 14.3	B → B	129
WB	207	0	207	0.29 → 0.29	13.3 → 13.3	B → B	97

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	1	100%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

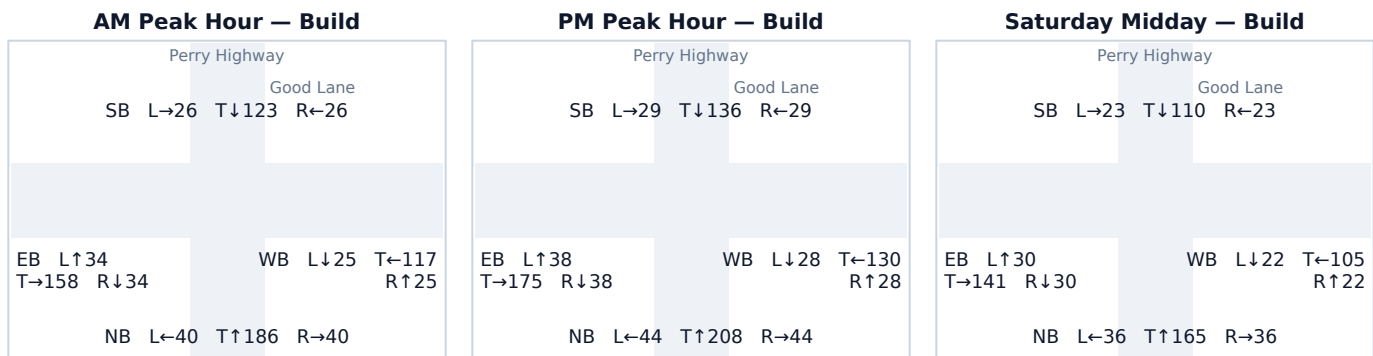
A.35 Perry Highway & Good Lane

Signal ID	pittsburgh-2226
Location	NW Pittsburgh · 1.56 mi from site
Added PM peak trips	< 1 (negligible)
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.51
Build (PM)	LOS B · 14.1 s/veh · v/c 0.51
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

The development distributes fewer than one net PM peak car trip to this junction, so the Existing (No-Build) and Build conditions are numerically identical at reporting precision. The junction is reproduced here for completeness; the scheme's net car-mode trip generation is below the level at which junction capacity governs.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	296	0	296	0.36 → 0.36	12.1 → 12.1	B → B	134
SB	194	< 1	194	0.24 → 0.24	10.8 → 10.8	B → B	82
EB	251	< 1	251	0.36 → 0.36	14.8 → 14.8	B → B	124
WB	186	< 1	186	0.27 → 0.27	13.7 → 13.7	B → B	88

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

A.36 Perry Highway & Lindley Lane

Signal ID	pittsburgh-2222
Location	NW Pittsburgh · 1.63 mi from site
Added PM peak trips	< 1 (negligible)
Existing / No-Build (PM)	LOS B · 15.1 s/veh · v/c 0.53
Build (PM)	LOS B · 15.1 s/veh · v/c 0.53
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.43 / EW 0.40; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

The development distributes fewer than one net PM peak car trip to this junction, so the Existing (No-Build) and Build conditions are numerically identical at reporting precision. The junction is reproduced here for completeness; the scheme's net car-mode trip generation is below the level at which junction capacity governs.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	279	0	279	0.36 → 0.36	12.6 → 12.6	B → B	129
SB	221	0	221	0.28 → 0.28	11.9 → 11.9	B → B	98
EB	257	0	257	0.36 → 0.36	14.0 → 14.0	B → B	124
WB	169	< 1	169	0.23 → 0.23	12.7 → 12.7	B → B	77

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.